Engineering Chemistry · Unit_01 · Molecular spectroscopy, phase equilibria, and computational chemistry

DRAFT — UE26CY151A · Unit_01 — Molecular spectroscopy, phase equilibria, and computational chemistry · Complete Unit Master

Notes-first study master · source-cited lecture explanations and compact page takeaways

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the edited lesson notes by topic. Evidence chips point to transcript intervals and slide pages. The complete source transcripts, page text, formula candidates, and image locators are preserved in sibling JSON datasets. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Unit_01 · Molecular spectroscopy, phase equilibria, and computational chemistry

1 UE26CY151A · Unit_01: Molecular spectroscopy, phase equilibria, and computational chemistry

This unit master organizes source material for Engineering Chemistry, Unit_01, officially titled Molecular spectroscopy, phase equilibria, and computational chemistry. It includes 14 video transcripts and 12 source slide decks, with page-level text and rendered images where available. This draft preserves source wording and page order; it does not claim a complete human subject review or full audiovisual review. EV0000001

SHORT-NOTE RECAP

Unit map · UE26CY151A · Unit_01: Molecular spectroscopy, phase equilibria, and computational chemistry

Official catalogue entry: UE26CY151A, Unit_01 (Molecular spectroscopy, phase equilibria, and computational chemistry). Bucket folder present: yes. Videos: 14; slide decks: 12.

EV0000001

2 Interaction of radiation with matter

Study focus: Interaction of radiation with matter. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000002 · EV0000003

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

I'm Dr.

Key details

- Lata Pasukuleti and I'll be teaching you this course on engineering chemistry.
- The content is like this.

- First we are going to look at interaction of electromagnetic radiation with matter, then electromagnetic spectrum, then we discuss the Born Oppenheimer approximation.
- Further we study BL Ambert's law and then we look at different types of spectroscopy.
- First we'll study microwave spectroscopy, then we go to IR spectroscopy and finally to electronic spectroscopy.
- In today's class, we will look at the definition of molecular spectroscopy, what we understand by interaction of electromagnetic radiation with matter, comparison between atomic and molecular spectrum.
- The quantization of energy concept, absorption and emission spectrum, and finally about spectroscopic units.
- The word spectroscopy has the in it spectrum.
- to study molecules.
- This happens because whenever we have a sample, say like this, a sample.
- And then we have electromagnetic radiation falling on this sample.
- The electromagnetic radiation has energy and this radiation is going to interact with the matter here.
- Now because of this interaction, there are bound to be changes in the electromagnetic radiation.
- Some of the energy could be absorbed, some of it could be transmitted, and some of it could get scattered.
- Nature of the matter.
- Here we are talking of molecular spectroscopy, so we are going to study how it is affecting the molecules.
- This interaction of electromagnetic radiation with matter is the concept which we are going to use in molecular spectroscopy.
- I need incident radiation.
- These radiation and this radiation is going to fall on the matter here, the molecules here, and there is going to be interaction.
- For that I need a detector or an analyzer we call it.
- Going to send this radiation to a diffraction grading.
- Because it's going to break the radiation into the different components of different wavelengths.
- And then we need a recorder which is going to record these different wavelengths, and what we get is.
- Known as a spectrum.
- So that means if radiation falls on a sample and we are able to analyze the radiation which is coming out, then we get what we call as a spectrum.

Example or application from the lecture

- So first let us define what is molecular spectroscopy.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000002 · EV0000003 · EV0000004 · +1 source units in pack.json

Core explanation · 05:00–09:59

It is a stream of particles which are called.

Key details

- Photons and these are moving through space in the form of waves.
- The radiation is going to have particle, we will see particle nature as well as wave nature in the form of we can call it as a stream of particles.
- particles called photons or we can look at it as a form of a wave.
- Radiations and they are oscillating perpendicular to each other and also perpendicular to the direction of propagation.
- electric field oscillating and we also have the magnetic field which is perpendicular and these are going to be perpendicular to the direction of propagation we can see in the picture over here and all electromagnetic radiations always travel with the velocity of light.
- These are important features which we should remember when we are studying molecular spectroscopy.
- C, what are the characteristics of the wave or how is a wave characterized?
- What are the various properties that a wave possesses?
- distance between two successive crests and troughs of the wave and it is it is denoted by lambda and it could be in meters, nanometers or angstrom depending on what is the distance between two successive casts or traps.
- Then we also have frequency.
- You can make out from the unit itself.
- minus thirty four Joues second and nu is the frequency.
- And wavelength is related to frequency by C is equal to nu lambda, c is the speed of light, lambda is the wavelength, and nu is the frequency.
- And there is also wave number which is related to.
- Wavelength by nu bar is equal to 1 by lambda.
- That's exactly what I said.
- All these properties, wave properties we should understand.
- Atomic spectrum is something which you have already studied.
- See, we are going to learn about molecular spectrum, but if you recollect what you studied in atomic spectrum, then understanding molecular spectr would be that much more simple.
- And these are the atomic orbitals which are present around the atom.
- This has just one electron in it, and we know that when radiation is going to fall on.
- This hydrogen atom, then these elect this electron is going to be excited to different levels.
- Which level it goes to depends on what is the amount of energy it absorbs.
- From one to three, one to four.

Example or application from the lecture

- What is frequency the number of cycles completed in a certain amount of time and that is given by nu, and it could be in hertz or the second inverse.
- We also have wave number for every wave and that is number of complete waves or cycles that is contained in a unit distance and that is given by new bar and that is centimeter inverse.

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EV0000006 · EV0000007 · EV0000008 · +1 source units in pack.json

Core explanation · 10:00–14:59

In to any of these higher energy orbitals, the electron transits when it absorbs radiation.

Key details

- In this case, it is atom.
- If we can record, we will get a spectrum like this.
- One, two, three, third level.
- Some will come to the first level.
- Some will come back to the second level.
- And the difference in energy for each of this.
- When it comes from here to here energy is different.
- If energy is different, the frequency and the corresponding wavelength are also different.
- Based on this difference in energy, we will get different lines in the spectrum.
- Electron coming back from the higher energy level to the lower energy level.
- One thing we can understand here that if we see a line in the spectrum.
- This is something you're physically observing.
- So if we see a line in the spectrum, it means that corresponding to that frequency, there is an energy gap which is present in the.
- In the atom, and that energy gap is between atomic orbitals because we are talking of atoms, it is between the atomic orbitals.
- The gap is equivalent to the energy which is getting emitted.
- In this case, this is an emission spectrum.
- So if you see a line in the spectrum, it means that there exist that two energy levels of that energy gap.
- That is about atomic spectrum and this spectrum is always is known as the line spectrum because you will see different lines corresponding to the energy emission which is happening here.
- Just like you have atomic orbitals, you have studied molecular orbitals also.
- Molecular orbitals exist inside the molecule just like in an atom there are atomic orbitals.
- That is known as the Homo, highest occupied molecular orbital.
- The electron now is going to transit from the highest or occupied molecular orbital to lowest unoccupied molecular orbital.
- And when such a transition happens, then we will see a line in the spectrum.
- That is because.
- Energy gap is there, that much energy is absorbed, say we are looking at absorption, then we'll find that particular line in the energy is the line corresponding to that particular energy is missing in the spectrum.
- The molecule above.
- Absorb that energy and transit it to the next level, and that is known as the lowest unoccupied molecular orbital.
- And this transition is to the lowest unoccupied molecular orbital.

- This is what we are going to see in our molecule.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000010 · EV0000011 · EV0000012 · +1 source units in pack.json

Core explanation · 15:00–19:59

We should see a line in the spectrum, right?

Key details

- But we find that in the case of molecules, we don't see a line.
- In fact, we see a band like this.
- If you look closely, if you have good resolution in your spectral photometer, you will find that these this band is actually made of several lines inside.
- What is this band spectrum in a molecule?
- We understand that in a molecule apart from electronic transition there could be other types of transitions possible.
- Because a molecule can undergo vibration.
- It can also undergo rotational rotation.
- So, therefore, apart from the electronic transition, there could be other transitions to different levels, like there could be change in vibrational levels, there could be change in rotation levels.
- An atom you have only atomic orbital, so you will see only electronic transition.
- There's only one line which you will see you will see only lines in the atomic spectrum, but in a molecule, you are going to see a band, and if you have good resolution, you will see finer lines inside this band.
- As I mentioned, if you see a line in the spectrum, that means there are corresponding energy levels which are equal to the energy gap.
- The frequency of the absorption or the emission which you see in the spectrum.
- Therefore, what do we understand so far?
- That atomic spectrum will give you a line spectrum, whereas molecular spectrum will give you a.
- molecular spectrum will be known as a band spectrum.
- You will see a band with finer lines.
- We know that energy levels in an atom or a molecule are not continuous, instead they are discrete or quantized.
- What do peep what do we mean by that?
- that we have energy levels at certain values only.
- It's not a continuous energy that is available inside the atom or a molecule.
- Like in a ladder you have rungs, different rungs.
- I you can put your hand leg on one rung or the next rung and the next rung.
- You cannot stand if you put your leg in between these rung.
- Exactly like that in an atom, you have these orbitals energy levels in a molecule also energy levels which are at discrete energy value.
- And then such an energy level is known as a quantized energy level.

- It is not continuous, it is quantized.
- Now because it is quantized, that means it will there will be a level here, there will be a level here.
- There cannot be any levels in between.
- Therefore, can be absorbed only in the form of packets which are known as quanta.
- By H nu.
- H is the Planck's constant and mu is the frequency of the radiation which will get absorbed or emitted.
- So that means energy is quantized means this h nu can be.
- Either as only H new or it could be two H new, three H new, four H new, but you cannot have a quarter H nu or a th three by fourth H new.
- No, that is what we mean by energy is quantities means the energy gap is always equal to some integral multiple.
- Of H nu.
- to this energy level and this difference in radiation is H nu.

Example or application from the lecture

- You can see this is another example where you're actually seeing a band.
- This energy gap is will be suppose there is an energy level E1 and there's an energy level E2, the energy gap delta E is equal to E2 minus E1.
- This difference in energy will be what we say quantize and therefore that is given.

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EV0000014 · EV0000015 · EV0000016 · +1 source units in pack.json

Core explanation · 20:00–24:59

Corresponding to the energy gap, the frequency of light is going to be absorbed.

Key details

- One is absorption spectrum, the other one is emission spectrum.
- What would be an absorption spectrum when light is absorbed.
- This would give you an absorption spectrum.
- The lower energy level to higher energy level, thereby absorbing a particular radiation.
- Absorbing a particular radiation would be like you will see some lines if you have used white light, then you would see some lines missing.
- That means there is a corresponding energy, two energy levels with that corresponding energy gap.
- That is known as an absorption spectrum, you can see here.
- And if the light has already been absorbed and now it is it's coming the atom or molecule is coming from the higher energy level to the lower energy level then what we see is known as the emission spectrum.
- So basically what it means that you can up it depends on the detector whether you're studying the absorption spectrum or you're studying the emission spectrum.
- Again, it tells us that based on the energy gap, light is going to be absorbed or emitted.

- It depends on what we are what is the detector, how we are analyzing or what spectrum we want to study.
- And most of our study here, we will be looking at absorption spectrum.
- Is expressed in joules.
- We know this relation also.
- Now, since c is equal to nu lambda, so nu bar can be expressed as nu by c, and therefore energy can be expressed in terms of nu bar.
- as HC nu bar.
- Why do we do that?
- Because new bar is treated as the spectroscopic unit of energy and basically it just tells us it corresponds to energy of a radiation and it is in terms of centimeter inverse.
- If we express why should we express in centimeter inverse and not in joules?
- It is just a matter of convenience.
- Because if you see you can try to do this conversion, one centimeter inverse is equal to 1.99 into 10 to the power of 23 joules.
- Minus 23 joules.
- So therefore 1 centimeter inverse is 1.99 tet power minus 23 joules.
- Amount of energy.
- Basically, we don't want to use such exponential terms, large exponential terms.
- Therefore, we can express in a simple way, we just say one centimeter inverse, five centimeter inverse, it's just a matter of convenience.
- What it means is this energy in joules.
- Has been expressed in terms of centimeter inverse.
- Though we know centimeter inverse is not an SI unit, but still it is for convenience that we will express energy in terms of centimeter inverse, and therefore it is referred to as the spectroscopic unit for energy.
- And before that, let me tell you that in the next class we are going to study about electromagnetic spectrum.

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EV0000018 · EV0000019 · EV0000020 · +1 source units in pack.json

3 Electromagnetic spectrum

Study focus: Electromagnetic spectrum. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000022 · EV0000023

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module one on molecular spectroscopy and in today's class we are going to look at electromagnetic spectrum and the information obtained from each region of the spectrum.

Key details

- Light is a form of electromagnetic radiation, which is divided into different regions based on the frequency of the radiation.
- Though these demarcations are made, the boundaries are not very precise.
- Then we have infrared radiation, then we have the visible region, then we go on to ultraviolet region.
- Then more energetic X rays and finally the gamma rays which are high energetic.
- First, the lowest energy beam, that is the radio frequency region.
- The wavelength is ten meters to one centimeter.
- That clearly conveys that the energy of this radiation is very low.
- It is known as NMR and ESR.
- NMR stands for nuclear magnetic resonance and ESR stands for electron spin resonance.
- Electron spin resonance is also referred to as electron paramagnetic resonance sometimes.
- What exactly do happens when radio frequency region light falls onto the sample.
- In the case of NMR that is called that is nuclear magnetic resonance there's going to be nuclear spin reversal, whereas in the case of ESR there's going to be electron spin reversal.
- We will start with electron spin because that is something you already understand that we can have two spins that is plus half and minus half.
- And both of them occupy the same orbital.
- So that means they are degenerate.
- Both these spins, plus and minus half, are degenerate and they occupy the same orbital.
- But in the presence of magnetic field, what happens is these will split into two levels one will be the lower level that is say minus half and the other one will be plus half.
- What is the difference?
- Say in one case it is the spin is clockwise and in the other case the spin is anti-clockwise.
- When you have a magnetic field, when these electrons are placed in a magnetic field, we find that while the energy levels now split into one with lower energy and one with higher energy, it is basically.
- How did this happen?
- Because the electron is spinning and it interacts with the magnetic field.
- One could be aligned with the magnetic field, the other could be opposite in direction to the magnetic field.
- Therefore, we get two energy levels now.

Example or application from the lecture

- Let us see what is the information we obtain from the different regions of the electromagnetic spectrum.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000022 · EV0000023 · EV0000024 · +1 source units in pack.json

Core explanation · 05:00–09:59

When a radiation falls on this atom, then we will find that if the energy gap is equivalent to the frequency of the radiation which is falling on this atom.

Key details

- then there will be a transition of the atom from one energy level to the next energy level.
- It is going to absorb this radiation and move from one level to the ne to from the lower level to the higher energy level.
- And what has happened because of this movement?
- There is a spin reversal.
- And why are we calling it as a resonance phenomenon?
- The energy gap.
- The energy gap means the frequency matches with the frequency of the radiation we know resonance will happen and the atom now picks up this energy and goes from the lower energy level to the higher energy level.
- But when there is a magnetic field present, then there is splitting, just like it happened in the electron.
- For the electron, the same phenomena happens for nuclei also.
- The spins are now you have one where it is aligned with the magnetic field and the other one where it is opposite to the magnetic field.
- And now we have two energy levels.
- When radiation is made to fall on these nuclei, then they are going whenever the energy gap matches with the incident.
- light frequency, incident radiation frequency, the atom now the nuclei are going to absorb that energy.
- There will be a resonance phenomenon.
- The nuclei will absorb energy and go to the next level.
- Again what happened?
- What we observe here is nuclear spin reversal.
- In the case of nuclei this would be plus my plus half and this would be minus half.
- that the energy gap is not very large.
- Therefore, a low frequency beam like the radio frequency region is sufficient to cause the spin reversal of either the nuclei or the electron based on what is the spectroscopy we are studying.
- Now here the wavelength is 1 centimeter to 100 micrometers means it is slightly more energetic beam and the spectroscopy studied in this region is rotational spectroscopy and we talk of rotational levels.
- We come to know of rotational levels.
- When the molecule rotates.
- There is a change in the dipole moment.
- We know that XL is a polar molecule.
- There is a change in the dipole moment.
- So as the molecule rotates, there is bound to be the fluctuation in the electric field because of the change in dipole moment.
- with which the electric field is fluctuating.

- That also has a fluctuating electric field because it's an electromagnetic radiation, there's a fluctuating electric field and a fluctuating magnetic field.
- Fluctuating electric field of the molecule due to rotation matches with the fluctuating electric field of the radiation falling on the molecule, resonance phenomena will happen, and the molecule now will move from.

Example or application from the lecture

- Now let us come to the next region that is the microwave region.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000026 · EV0000027 · EV0000028 · +1 source units in pack.json

Core explanation · 10:00–14:59

one rotational level to the next rotational level.

Key details

- Because it is by virtue of the rotation of the molecule.
- If we have homonuclear molecules like say hydrogen, then we find that we cannot study rotational spectroscopy.
- Because even when the molecule rotates, there is no rotating electric field, and if you drop radiation.
- the incident radiation onto this molecule, we will find that there will be no interaction as there is no fluctuating electric field.
- And the study we do here is known as vibrational spectroscopy, and now we will learn about vibrational levels.
- When a molecule undergoes vibration, say a diatomic molecule, simple diatomic molecule, when it undergoes vibration, again, say it is stretching.
- The frequency of the elect fluctuating electric field due to vibration of the molecule, the molecule is going to absorb this radiation, there will be a resonance like phenomena.
- Molecule will absorb radiation and it will move to the next level.
- We call it called as the vibrational levels.
- Because it is due to the vibrational motion of the molecule.
- One thing we can see very clearly here that rotational transition requires the.
- Radiation to be in the microwave region and whereas for vibrational transitions the we require infrared region which is much more energetic.
- Therefore, it requires infrared region, whereas in the microwave region they are closer as it the transition requires only the less energetic beam as compared to infrared.
- It's only when you come to triatomic molecules you can notice the bending motion also.
- The dipole moment, hence no change in electric field.
- So therefore no fluctuating electric field and there will be no interaction with the incident radiation.
- And therefore, we say that the homonuclear diatomic molecules again are not active in the infrared region.

- We know visible comes first and then the UV region, but nevertheless the spectrum with the spectroscopy is referred to as UV visible spectroscopy and the wavelength now has become one micrometer to ten nanometers.
- Here now we are going to look at electronic states.
- Because we understand that the molecule or the atom is going to absorb energy and.

Example or application from the lecture

- Let us get to the next beam of next region that is the visible and the UV region.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000030 · EV0000031 · EV0000032 · +1 source units in pack.json

Core explanation · 15:00–19:59

the m atom will go to the from the electron now transits from one atomic level to the next atomic level.

Key details

- Similarly in a molecule it goes from one molecular orbital to the next molecular orbital.
- There are electronic transitions happening and that will happen between the different electronic states.
- We must understand it's only from the Homo to the lumon, that is the highest occupied molecular orbitals in the case of molecules.
- That will be the valence electrons, the far the higher energy electr electrons because they require the minimum energy t the energy is required to move them is much less as compared to the one the an electrons which are closer to the nuclei as in the case of atoms and in this case in the molecule also it will move from Homo to lumo.
- Therefore there is change in electronic disposal.
- Distribution of the valence electrons.
- U if it happens in the visible region we can see the color.
- If it's in the UV region, obviously we cannot see the colour.
- These are very energetic beams and now the wavelength has come down to picometers, so 10 nanometers to 100 picometers, and this study is known as X-ray spectroscopy.
- But these transitions now will be involving the inner electrons.
- In the case of UV visible, it is the valence electrons which undergo transition and that requires much less energy.
- Whereas if you want to remove the inner electrons then it requires more energy and therefore x-ray region is what is required the light in the X ray region, the frequency should be high, energy should be more to cause this electronic distribution which involves the inner electrons.
- We are aware that gamma rays are produced whenever there is nuclear reactions happening.
- We know that nuclei consists of protons and neutrons which are very tightly held.
- we have to use very highly energetic beams in order to cause any nuclear rearrangement.

Example or application from the lecture

- We know that whenever the radioactive nuclei changes into another nuclei, gamma rays are given out.
- In all the other cases we are concentrating more on the absorption spectroscopy, but here we will study more it's the emission spectroscopy and based on the gamma rate which is being given out, we can get information about what in nuclear rearrangement happened and what were the nuclei which are getting formed or what is the nuclear reaction which is happening.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000034 · EV0000035 · EV0000036 · +1 source units in pack.json

Core explanation · 20:00–24:59

is giving us different information about the atom or the molecule which is being impinged upon by the radiation.

Key details

- One more type of spectroscopy which is very important is Raman spectroscopy.
- But it is different from all these like rotational or vibrational spectroscopy.
- or electronic spectroscopy.
- Because there we talk about absorption of light, there or emission of light.
- But here Raman spectroscopy talks about scattering of light.
- Radiation will get transmitted.
- Most of the radiation we expect apart from what is absorbed is going to be transmitted.
- transmitted radiation, but some amount of radiation there will be a change in direction and this is referred to as the scattered radiation.
- Normally in transparent medium we expect very little scattering, but nevertheless there is a possibility that scattering will happen.
- Whatever incident radiation is there, the same wavelength will be seen most of the time for the scattered radiation and such a scattering is referred to as.
- Ra r rarely scattering.
- But Rama noticed that some of the scattered radiation consists of radiation of a different wavelength from that of the incident wavelength.
- Most of the times the scattering which happens is going to be rally scattering where the incident wavelength and the scattered the incident light wavelength and the scattered light wavelength has the has is same.
- There will be some radiations which we find that the wavelength is more.
- of the scattered radiation which we refer to as the Stokes Raman scattering or they when you see in the spectrum they are called as Stokes lines and some of the radiation we will find will have a smaller wavelength and those are referred to as the anti scale.
- Stokes, Raman, scattering, and in the spectrum they are referred to as the anti-stokes lines.
- And that these photons carry.
- E is equal to H nu.
- We know that E is equal to H nu.
- And again it cannot remain there because it's an unstable state from the ground state.

- From the ground state it goes to a virtual state and then it comes back with the same frequency of radiation which is referred to as elastic scattering.
- same kind the same frequency of radiation and this is referred to as valley scattering or this is elastic in nature, elastic scattering.
- But there could be what Raman observed, but there were some radiations which were inelastic scatter were inelastic.
- scattering was happening.

Example or application from the lecture

- Let us see what are these Stokes and Antistokes like now we can e we can explain this quantum mechanically if you consider light to be made up of photons.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000038 · EV0000039 · EV0000040 · +1 source units in pack.json

Core explanation · 25:00–29:59

initial ground state it ends up at a higher level and therefore now the frequency is much less.

Key details

- See to a higher level, therefore the energy of this beam is much less.
- It is now you have to subtract this much of energy.
- It is not at the ground level, it is starting at a higher level, it goes up.
- But when it comes back, then there the it comes back to the ground state.
- That means now the energy is more.
- You can see the energy gap, it is much more and therefore now the beam has more energy or smaller wavelength, and this is known as anti stokes scattering.
- And we find that.
- Whenever this scattering happens, inelastic scattering, there could be you will see stokes lines as well as anti-stokes lines.
- Of course, it is found that the stokes lines are much more intense as compared to the anti-stokes lines.
- That is because the population of the ground state will be any time more than that of the higher level.
- So anti-stokes lines will appear only if there are molecules in this higher level they go absorb energy and then come back to the initial ground state and therefore their energy they are much more.
- And one advantage of Raman spectroscopy is that homonuclear diatomic molecules I mentioned just now will not show any absorption in the microwave or the.
- all the information which we couldn't get from the pure rotational or pure vibrational spectroscopy we can obtain from Raman spectroscopy.
- And of course we can see as we go from one region to the other, the frequency is increasing, the wave number is decreasing.
- Length is decreasing and the wave number is increasing.

- So in the next class we are going to talk about Born Oppenheimer approximation and then we will also move to rotational spectroscopy.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000042 · EV0000043 · EV0000044 · +1 source units in pack.json

4 Rotational spectroscopy

Study focus: Rotational spectroscopy. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000046 · EV0000047

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We continue with module one on molecular spectroscopy and in today's class we are going to derive the expression for rotational energy levels of a diatomic molecule.

Key details

- We will study the selection rules and finally we will arrive at the rotational spectrum.
- We need a model in order to understand how the molecule is going to behave, and we can see here what we are going to use is a rigid rotor molecule.
- We have two atoms heteronuclear so the masses are different and they are held together by a rigid rod and that is having a center of gravity at the point C.
- which are held together by a rigid rod, meaning that whenever there is rotation, this rod will not change its form.
- Now we have masses M1 and M2 which are joined by a thin rod, so therefore R0 is going to be R1 plus R2.
- Therefore, we can write the balancing equation m1r1 is equal to m2r2.
- M2 R2 R1 plus M1 R1 R2.
- All we have done is replaced M1 R1 by M2 R2 and here M2 R2 by M1 R1.
- And what we get is I is equal to R1 R2 M1 plus M two.
- Because there is a rotation which is happening.
- Now since m1 r one is equal to m2 r2 we can also write this as m2 r naught minus r1 because we said that r naught is equal to r one plus r2.
- And why do we require this?
- Because we want to substitute R1 here.
- So therefore R2 is equal to M1 R0 by M1 plus M2.
- When we substitute for R1 and R2 in this expression for moment of inertia we get this expression i is equal to m1 m2 divided by m1 plus m2 r naught square which can be written as mu r naught square.

Example or application from the lecture

- So let us get to the expressions of rotational energy levels for a diatomic molecule, because that is the simplest molecule.
- So let us take the case of a diatomic molecule and heteronuclear diatomic molecule of course because homonuclear diatomic molecules will not be microwave active.
- Let us see how the derivation goes.

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EV0000046 · EV0000047 · EV0000048 · +1 source units in pack.json

Core explanation · 05:00–09:59

I is equal to mu r naught square.

Key details

- And what is r naught?
- It is the distance between the two atoms or the two masses which we have considered here.
- We are still talking in terms of classical mechanics.
- Energy is half I omega squared.
- Omega is the angular velocity.
- This can be written as L squared by 2i.
- Because L is the angular momentum.
- because we are talking of a single molecule and here classical mechanics or classical laws are not applicable to molecules which are so small.
- What does this expression show us that the angular momentums are all quantized.
- They are not continuous.
- In the classical case we said L squared by 2i.
- L can have any value, but here L cannot have all the values.
- It can take up only those values which J has because.
- Here do j plus 1 by 2 pi h.
- So therefore the quantity j is known as the rotational quantum number and it can take up integral values 0, 1, 2, 3.
- So we will get different values of quantum the angular momentum based on the value of j which we substitute here and therefore we can see that angular momentum is quantized.
- How we can substitute for L in this expression for energy, and what we obtain is EJ.
- is equal to h squared by 8 pi squared i j g plus 1 joules.
- This energy is expressed in terms of joules and what is h here it is the Planck's constant 6.626 into 10 power minus 34 joules second.
- And I is the moment of inertia of the.
- Which energy are we talking about?
- The rotational energy levels inside the molecules is R context.
- Because they cannot have any value but those which are decided by the value of the rotational quantum number that is J.
- And we know that J can take up only values of 0, 1, 2, 3.
- If I want to convert the energy into wave number we need to divide by H C.

- That was H squared by 8 pi squared I, so it becomes H by 8 pi squared I C, J times J plus 1 centimeter inverse.
- Now this is a constant for a particular molecule because H pi and C are constant C is the speed of light and that is 3 into 10 to the power of 8 meters per second and i is the moment of inertia of the molecule which is again constant.
- Therefore this can be written in the form of a constant b.

Example or application from the lecture

- So therefore we will use the Schrödinger equation for the rigid rotor, and then we find when we solve the Schrödinger equation that angular momentum is now given by this expression root of j plus 1 by 2 pi h or we can say root of j plus 1 h cross which can also be written as h by 2 pi.
- Let us see what these energy levels look like.

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EV0000050 · EV0000051 · EV0000052 · +1 source units in pack.json

Core explanation · 10:00–14:59

1 centimeter inverse.

Key details

- And what is the value of B is H over A pi squared I C centimeter inverse and B is known as the rotational constant.
- Is known as the rotational constant and it's expressed in centimeter inverse, it will be h by a pi squared i c.
- And that's how we arrived at the expression for the rotational energy levels.
- This is the energy scale.
- And we have 0 is at g is equal to 0, we have energy as 0.
- When we substitute j is equal to 1, we get 2b.
- And that's what is shown here.
- And when you get go to j is equal to 2, we get 6b.
- We can see this is 0, 2b, 6b and so on.
- We can go to the next level that is 12b.
- If you want to find out for any energy level, all you have to know is the value of B.
- When we substitute the values for J 0, 1 to 3 we get the different energy levels.
- But just knowing the rotational level is not going to help us get the spectrum because what we want finally or what we get when we do this spectroscopy is the spectrum.
- We have to follow certain rules which are known as the selection rules.
- If we understand the selection rules, then we will be able to arrive at the spectrum.
- The selection rule for rigid rotor model.
- Is obtained after solving the Schrödinger equation.
- But there is another specific selection rule which is employed and which is applicable only for a rigid rotor heteronuclear diatomic molecule and that says delta J is equal to plus minus one.

- When it is plus one we refer to absorption spectrum.
- When it is minus one we refer to the emission spectrum.
- We know is equal to the rotational levels where are they?
- Bj plus 1 centimeter inverse.
- We have these energy levels inside the molecule.
- Just now we saw.
- Delta J is plus minus 1 means that.
- And we can have a transition zero to one, we can have a transition one to two, we can have a transition two to three, we can have a transition two to three to four, but we cannot have zero to two, two to four.
- All those transitions are forbidden.
- Only transitions allowed are between one level and the next level present in the molecule.
- Why am I interested in the energy gap?
- Because we know that the radiation will be absorbed based on the energy gap.
- Rotational levels, of course, we know where they are, but more important for us is to know what is the energy gap.
- What is the energy gap between J to J plus 1?
- So we substitute here J plus 1 first minus for so J.

Example or application from the lecture

- And these are rotational energy levels because we consider that the molecule is rotating.

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EV0000054 · EV0000055 · EV0000056 · +1 source units in pack.json

Core explanation · 15:00–19:59

wave number and that will be the expression becomes 2bj plus 1 centimeter inverse.

Key details

- It is going so J is equal to zero.
- That's what we mean.
- That is one.
- When we go to from one to two, it will be four b.
- When we go from two to three, it is going to be six b and so on so forth.
- We can calculate the energy.
- the energy gap between the energy levels and we see very clearly as we go to higher energy levels the energy gap is going on increasing.
- And if you have a transit and only delta j is equal to plus minus 1 is what is allowed.
- And here we can see plus 1 is what we are doing here.
- Zero then one to two, then it's going to be four b and then six b and eight b and ten b so on so forth.
- the energy gap itself is going on increasing.
- Energy will be absorbed.
- And what energy will be absorbed?

- That will be corresponding to the energy gap.
- So that means when radiation falls on this molecule, the first line will appear at 2b because that is the lowest energy gap, and remember this is in centimeter inverse this way.
- That means wave numbers energy is increasing.
- You can see from the energy level diagram here that this requires minimum energy.
- Energy and that will correspond to 2B.
- 0, 2, 1.
- Then we have 1, 2, 2, that would be 4b, and then we have 6b here, and then we have 8b, 10 P and 12B.
- The first line is appearing at 2b, but.
- The difference between these lines is also very clear.
- It is going to be 2b.
- of H C.
- So you will have the first line appearing at 2b.
- And the distance between any two lines in the spectrum is going to also be equal to 2b, and that remains constant.
- And from here we can calculate what is to be.
- Mu is the reduced mass which is fixed for any molecule which is mu is equal to we already know m1 m2 by m1 plus m2.
- And what is R0 here?
- What was our model?
- This is R0 and what is that?
- Will be for any molecule it is going to be the bond length.
- So that means.

Example or application from the lecture

- Let us see the spectrum.

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EV0000058 · EV0000059 · EV0000060 · +1 source units in pack.json

Core explanation · 20:00–24:59

Using I the microwave spectroscopy or rotational spectroscopy, we can determine the bond length of the molecule.

Key details

- You can see that the intensity of the lines is increasing and then.
- Intensity depends on the population of the energy levels and therefore we can see from this spectrum that it is not just the ground level which is the most populated.
- Why we are saying that because this refers to zero to one transition, but this is not.
- The most populated level at room temperature, even some of the higher levels are also populated.
- It is not just the lowest level or the ground state which is the most populated in the case of rotational levels, even some of the higher levels are also populated, as you can see from the

intensity of the lines.
- So in the next class we are going to study another type of spectroscopy, which is the vibrational spectroscopy.

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EV0000062 · EV0000063 · EV0000064 · +1 source units in pack.json

5 Vibrational spectroscopy(harmonic oscillator)

Study focus: Vibrational spectroscopy(harmonic oscillator). This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000066 · EV0000067

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module one on molecular spectroscopy.

Key details

- And in today's class we are going to study about vibrational spectroscopy.
- We will tie try to derive an expression for the vibrational energy levels that is for a harmonic oscillator and also we will arrive at the vibrational spectrum for a harmonic oscillator.
- Like I have explained this before when we were talking about the electromagnetic spectrum.
- It is vibrating.
- And then compresses stretches and compresses that would result in the a vibrational change in the molecule.
- Now when there is a vibration in the molecule, then the dipole moment is also changing because we know HCl is a polar molecule you have a slight positive charge and a slight negative charge.
- The direction remains same but same but the magnitude is changing.
- Field is also changing.
- The electric field now is there is a frequency with which it is vibrating, and therefore the electric field will also fluctuate with a particular frequency.
- We know that electromagnetic radiation it has an electric field as well as a magnetic field and based on the energy of the radiation, the electric component of the electromagnetic radiation is also going to be fluctuating.
- One vibrational level to the next vibration level.
- Rotational spectroscopy we know that we use the microwave energy.
- But here we are going to use the IR radiation.
- That means that the vibrational energy levels, the energy gap of these vibrational energy levels matches with the energy gap of the IR radiation.
- Because only when these frequencies there is going to be a resonance phenomena and the molecule goes from the lower vibrational level to the next vibrational level.

- So what we see as a result of this transition, because there is a transition which is happening from the of the molecule from lower vibration level to the higher vibration levels, we will see that a particular energy is being absorbed from that whole IR region beam.
- So therefore that would if we can record.
- Is that energy that is getting absorbed?
- Then we can get what we call as the vibrational spectrum.
- And once we have the vibrational spectrum with us, we will be able to understand about the nature of the molecule and even.
- Then we can identify what is the molecule.
- Therefore, this vibrational spectroscopy is like a fingerprint for the molecule.
- These are examples.
- Why these examples?

Example or application from the lecture

- Suppose we have an H Cl molecule and it is going to undergo.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000066 · EV0000067 · EV0000068 · +1 source units in pack.json

Core explanation · 05:00–09:59

They there when the vibration takes place there has to be a change in the dipole moment.

Key details

- So therefore these molecules we know are polar in nature and therefore we will find that they will vibr when they vibrate there is going to be a change in the dipole moment.
- But when we.
- Come to homonuclear diatomic molecules like H2 and Cl2, we observe that these will not be active in the IR region.
- Because they do not possess a dipole moment.
- They are found to be I inactive.
- Can we say that if a molecule possesses permanent dichole moment, then it will be active in the IR region, just like we said about rotational spectroscopy?
- Here there is a small difference that CO2 molecule, we know that.
- It is microwave inactive because it does not possess a permanent dipole moment.
- But here in this case, in the IR region, we find that CO2 molecule is actually IR active.
- The reason is.
- Here, the criteria for observing a transition between the vibration levels is that.
- When the molecule is vibrating, there should be a change in the dipole moment, and that happens in the case of the CO2 molecule.
- Though it doesn't possess a permanent dipole moment, we know that the CO2 molecule, each of these bonds possess a dipole moment which actually cancel against each other, but when it vibrates we will find that in certain modes of vibration the CO2 molecule we can actually see a change in the dipole moment.

- But if the molecule is undergoing asymmetrical stretching, that is you can see one bond is becoming small, one bond is stretching, and then it goes in the reverse direction.
- If this happens, then we will find that there is definitely a change in the type moment because these two moments are not getting cancelled.
- So therefore we find that in this mode it becomes IR active.
- Because there is a change in the diple moment.
- And then if it is bending, say in the plane it is bending like this, again we see there is a change in the diple moment, these movements are not getting cancelled now, and therefore in the bending mode also we find that it is I active.
- And we will arrive at the energy expressions and see.
- why is the molecule able to show the vibrational spectrum.
- And again as we did in rotational spectroscopy we need a model so that we can apply that model and get to the energy expression.
- We are taking a heteronuclear diatomic molecule, so one mass will be larger than the other.
- And now we have a spring which is attaching the two atoms or holding the two atoms.
- So therefore when the molecule stretches the spring is getting stretched, when it is compressing the spring is getting compressed, and in the equilibrium position it remains like this.

Example or application from the lecture

- in the case of symmetric stretch, now suppose the CO2 is stretching symmetrically about the center then we about its equilibrium position we will find that there is no change in the dipole moment at all.

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EV0000070 · EV0000071 · EV0000072 · +1 source units in pack.json

Core explanation · 10:00–14:59

And since we it is a simple harmonic motion that we observe and it's a simple harmonic oscillator we can say that the more this vibration is going to obey hook's law.

Key details

- These molecules are never at rest.
- These bonds are always vibrating in different ways.
- Therefore, if we can get the IR spectrum, then we will get a fingerprint of the molecules.
- And if that will be very unique to the molecule.
- It obeys Hook's law, simple harmonic oscillator, f is equal to minus of kx.
- Any displacement is now related to the force by minus of kx.
- R minus r equilibrium will be the displacement.
- This means that K would basically be F.
- By X, ignoring the negative sign because it's just because it's a restoring force.
- So if the spring constant or the force constant is very high, that means even for a small distance.
- We need to apply a large amount of force.

- And looking at this expression we can understand that the energy curve is going to be parabolic.
- what is the reason why we get energy profile like this?
- As a rule, we understand that the heavier atoms we are going to fix at a particular position and it is the lighter atom which is going to be moving when the molecule stretches.
- That means we are trying to expand it.
- H now we are stretching.
- Earlier this was the bond length, you are providing more energy, you are stretching it more, so the bond length increased and it will reach this position, and when you compress it, it comes back to this position again, both are of higher energy.
- Every one has a.

Example or application from the lecture

- So let us say that the Cl is fixed because being the heavier and since it is at equilibrium so at this is the equilibrium distance or the internuclear distance at Cl that is in the equilibrium position.

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EV0000074 · EV0000075 · EV0000076 · +1 source units in pack.json

Core explanation · 15:00–19:59

M1 plus M2.

Key details

- So if we want to express in terms of wave number because we are interested in spectroscopic units, so nu bar oscillation would be 1 by 2 pi.
- C root of K by mu.
- 1 by 2 pi C root of K by mu centimeter inverse.
- Mu is a known quantity for a particular molecule.
- K, what is K?
- It is known as a force constant, but in the case of a molecule, what would be K?
- K would represent the bond strength.
- Then somehow, if I can find out what is the new bar oscillation for this harmonic oscillator, I will be able to get the value of K using this expression.
- Now what we do just like we did in the case of rotational spectroscopy we need to solve the Schrödinger equation because so far we are talking of classical mechanics but now we know we are talking of a molecule we cannot apply classical mechanic laws mechanics laws therefore we need to you go give it a quantum treatment.
- That is in Joule.
- We got a term V which is known as which is the vibrational quantum number and can take up values zero upwards.
- That is V is equal to 0, 1, 2, 3, and so on so forth.
- That means what this energy now expression because of the V has become quantized.
- But here now the energies are have become discrete or they have become quantized.

- And when we do it in wave numbers we know that it's represented by this small e and this is equal to v plus half nu bar centimeter inverse.
- So here it was V plus half h nu but because we are converting it dividing by H C, so this becomes nu bar centimeter.
- inverse so this is an important expression and this is going to give us the various energy levels when we start substituting v is equal to 0123.
- So we got that expression when we substitute V is equal to 0 because it was V plus half nu bar.
- If we substitute V is equal to 1, then we get 3 by 2 mu bar centimeter inverse.
- If you substitute 01 to V, I can get the vibrational levels in the molecule.
- What it looks like.
- The energy levels like this.
- In the classical mechanics, all these energies are allowed, but now we see that the energy of the quotient.
- What is the energy?
- Hal new bar.
- So that means it is not here, but it has a value that is half new bar, and therefore this energy is known as zero point energy.
- What does it mean that.
- The lowest energy level the lowest energy level is not at an ease equal to zero.
- We are not getting ease equal to zero, rather we are getting a value for this and that is half new bar.
- This is referred to as zero point energy.
- That is three by one.

Example or application from the lecture

- Let us see where these energy levels are appearing.
- Basically means that a molecule is never addressed, but it is always vibrating with energy which is known as zero energy and given by now let us start two B is equal to one.

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EV0000078 · EV0000079 · EV0000080 · +1 source units in pack.json

Core explanation · 20:00–24:59

two energy scale so let us go to the next one and that will be three by two new bar so now we have the next energy level now when we so this is zero this is one now let us say we have two here again this is five by two new bar and we have a four year save that will be three.

Key details

- And then it would be seven by two nubar.
- We get half nubar, three by two nubar, five by two nubar, and seven by two nubar.
- thing is we are interested in the energy gap because that will actually tell us the absorption that is going to take place.
- Between zero and one.

- What is the difference in energy?
- That would be B is equal to one.
- Minus V is equal to zero.
- V is equal to one is substitute energy expression.
- And then V is equal to zero is substrate the energy expression and V get the energy gathered.
- And that would be three by two minus half.
- new bar and that is equal to new bar.
- So therefore between these two energy levels 0 to 1, the energy gap is new bar.
- That is one to two.
- Again you are five by two minus three by two are so and again when you go from two to three again the energy differences so what do we understand here that energy levels.
- The quantity of energy levels difference is equal to nu bar oscillation centimeter inverse.
- where the energy levels are.
- You also know what is the energy gap.
- is still not sufficient for me to arrive at the spectrum.
- What is important, I should know what are the selection rules.
- When it remains the same that it is same for all molecules.
- That when a molecule vibrates there should be a change in the diagonal moment.
- And what is the specific selection rule for the simple harmonic oscillator diatomic molecule?
- It says that Dell B is equal to plus minus.
- What does it mean?
- That the only transition allowed will be zero to one or one to two or two to three.
- You cannot have zero to two or one to three and or any of them any other transition two to four, etc.
- Therefore we see that now zero to one where would the line appear.
- What do you mean by where would the line appear in the spectrum?
- That it is going to absorb that much energy equal to the energy gap that we have.
- From V is equal to one, V is equal to two, again the entity are the same.
- It is equal to number centimeter in now we can see from the spectrum here.
- from the rotation the vibrational levels you can see that the energy gap is constant.
- And this is.

Example or application from the lecture

- Let us go from 3 by 2 to 5 by 2.

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EV0000082 · EV0000083 · EV0000084 · +1 source units in pack.json

Core explanation · 25:00–29:59

being restributed by L V is equal to plus minus one.

Key details

- Therefore when you see the absorption you can only go to the next constituent level.

- Therefore the spectrum is going to look like this.
- Just one line is observed in the spectrum and that will be at new bar centimeter in.
- The other terms are known to me.
- from the spectrum.
- And this is going to.
- Help us find out what is the bond strength of the molecules.
- Rotational spectroscopy saw that it gives us the bone length and vibrational spectroscopy is going to give us the bone strength.
- So in the next class we are talking to talk about how this harmonic oscillator model is not suitable for molecules, and we will take up [ASR low confidence; check audio].
- Vibrational spectroscopy by changing the model to an end-harmonic oscillator.

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EV0000086 · EV0000087 · EV0000088 · +1 source units in pack.json

6 Vibrational spectroscopy(anharmonic oscillator)

Study focus: Vibrational spectroscopy(anharmonic oscillator). This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000090 · EV0000091

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We continue with module one on molecular spectroscopy, and in today's class we will look at the anharmonic oscillator model for vibrational spectroscopy.

Key details

- That's because we considered E is equal to half k x squared that is for a harmonic oscillator and that would result in the energy profile being that of a parabola.
- Like we can extend the bond length as much as we want, but still the molecule remains intact is what we see in this curve.
- But the real picture is something different.
- We know that if we stretch a molecule beyond a point, it is going to dissociate.
- So therefore we need to change the model for real molecules from simple harmonic oscillator to that of the anharmonic oscillator.
- As we come closer, we are reducing the bond length because.
- What is the bond length?
- This is the bond length.
- That is the equilibrium bond length.
- But when we come to this point where we have reached the equilibrium bond length of the molecule, then this is the most stable and therefore represented by minimum in the curve, minimum in the energy curve.

- Atoms even further closer, then we are compressing the molecule, and again the energy is going to increase.
- model using the anharmonic oscillator model.
- It says potential energy is equal to d equilibrium 1 minus exponential a times r naught minus r the whole square.
- Till there is zero, and you can very well make out that this would represent the dissociation energy of the molecule.

Example or application from the lecture

- Then we consider and derive the expression for vibrational energy levels for an anharmonic oscillator, and finally we will try to get the vibrational spectrum of an anharmonic oscillator.
- The reason being that when you consider it as a harmonic oscillator, we are assuming that we can go on stretching the molecule and it will remain intact.
- When we say the molecule behaves unharmonically or vibrates in an anharmonic fashion, then the best way to represent the energy profile is given by this curve which is shown in green.

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EV0000090 · EV0000091 · EV0000092 · +1 source units in pack.json

Core explanation · 05:00–09:59

And how do we get the energy level expression?

Key details

- So nu E bar because we are representing in terms of centimeter inverse.
- And what is V here?
- V is the vibrational quantum number.
- V is the vibrational quantum number which is going to take up the value 0, 1, 2, 3.
- And therefore by substituting these values of V, V can get the different energy levels which are present in the anharmonic oscillator.
- But only thing here we have some new term here.
- What is this new term?
- and whose value for bond stretching vibrations is always small and positive.
- We have to remember this.
- It is small and positive.
- But so the oscillation frequency expressed in wave number.
- What do these energy levels look like?
- All we have to do is substitute values for V is equal to 0, 1, 2, 3.
- And you can notice here that we have used the anharmonic oscillator model.
- To zero, then we are going to get an energy level here.
- When we substitute V is equal to one, obviously we got this energy level here.
- Again slightly lower than that of the harmonic oscillator.
- When we substitute V is equal to 2, there is again a decrease in energy because of the second term, but remember that this term is getting squared.

- So that means each time we are subtracting a larger number what does it mean V is equal to 2 level will be much more closer to V is equal to 1 as compared to in the harmonic oscillator model and similarly 3 4 5 6 as we go on substituting the.
- Energy levels are going to come closer and closer and we see that as we go to higher levels the vibrational energy levels are actually coming closer.

Example or application from the lecture

- Just like we did in rotational spectroscopy or when we were talking of harmonic oscillator, we have to solve the Schrodinger equation for the harmonic oscillator and the vibrational energy levels now will be given by this expression E that is we are now talking in centimeter inverse so that means the energy in joules divided by H.
- V plus half nu bar was the expression for a harmonic oscillator but here we have a second term because of the anharmonicity and that is given by minus of V plus half squared nu e bar X E.

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EV0000094 · EV0000095 · EV0000096 · +1 source units in pack.json

Core explanation · 10:00–14:59

What did we see in the case of rotational energy levels?

Key details

- That the energy levels are going farther and farther apart.
- When we came to harmonic oscillator, we saw that the energy gap remains constant.
- You go to higher levels of V also, energy gap remains constant.
- That's why we saw only one line in the spectrum.
- But here, what do we observe?
- Because of this second term getting subtraction.
- And larger and larger term is getting subtracted each time, we find that the energy levels are coming closer together.
- Here we can see that D equilibrium, that is the dissociation energy.
- And there's one more D naught over here.
- What would this be?
- This is the dissociation energy required by the molecule.
- What do we mean?
- Because we know that in the case of vibration,.
- A molecule is always going to possess the zero point energy.
- When a molecule vibrates there should be a change in the dipole moment.
- And what is the specific selection rule now?
- Delta V is equal to plus minus one, plus minus two, plus minus three, so on.
- All transitions are going to be allowed.
- In the case of harmonic oscillator, what was the selection rule?
- Only plus minus one.
- And all other transitions were forbidden.

- But here we have plus minus one, plus minus two, plus minus three, meaning that all the transitions are going to be permitted.
- E versus R.
- And now your energy profile is like this.
- So what it means, delta, and we know that the energy levels are going to be.
- Coming closer like this.
- Just a rough diagram.
- And what do we mean plus minus one is allowed?
- Minus will stand for emission.
- Plus minus one is allowed.
- Then plus minus.
- Two is also allowed.
- And again plus minus three is also allowed.
- So that means you will have a whole lot of transitions that are possible.
- And you can have 0 to 1, 0 to 2, 0 to 3, you can have 1 to 2, 1 to 3, 1 to 4.
- Two to one two to three, two to four, all these transitions are going to be allowed.
- It is going to be very, very complicated and it'll be difficult for us to interpret the spectrum.
- but we have something coming to help us.
- And what is that?
- Most of the transitions now start from V is equal to zero level.
- Because this is the state which is the most populated state.
- All of the transition, most of the transitions are going to start from V is equal to 0.
- Though the selection rule actually allows it.
- But now the spectrum is going to be much simpler.
- Because all transitions will start from these equal to zero level at room temperature when you're recording the spectrum.
- So first one will happen from V is equal to 0 to 1 and that is known as the fundamental absorption.
- Then second one will be 0 to 2, which is known as the first overtone, and then we have v is equal to 0 to 3, that is known as the second overtone.
- From V is equal to zero.
- So you will see in the spectrum fundamental absorption, first overturn, second overturn.
- You can have other transitions also two is equal to four or five, but then those will be very weak in intensity.
- In fact fundamental absorption will be the one which will be most intense and then we have first overtone and second overtone.
- After that the intensity of the transitions become very weak because more energy will be required.

Example or application from the lecture

- If you remember, let us draw this and see what it looks like.

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EV0000098 · EV0000099 · EV0000100 · +1 source units in pack.json

Core explanation · 15:00–19:59

Fundamental absorption would correspond to V is equal to 0 to V is equal to 1.

Key details

- To 1 and minus V is equal to 0, the expression we get is this mu E bar 1 minus 2 XC centimeter inverse.
- Remember that we are expressing energies in terms of wave numbers.
- Though I have written energy gap here, what do we mean that?
- You all should substitute for yourself and see whether you are getting this expression.
- Go to the energy expression, substitute V is equal to 2 minus V is equal to 0, and this is the expression you get for the first overtone.
- And similarly, second overtone, you.
- Going to get an expression like this.
- We know that the line in the spectrum you're going to see.
- is based on the energy gap because that is the amount of energy the molecule is going to absorb to go from one level to the next level.
- If it was.
- A harmonic oscillator and then we would have seen the line at nu bar, but now there is a 1 minus 2 XC clearly indicating that the energy gap is now less.
- allowed us a plus two and similarly three mue bar if it allowed us a plus three.
- We will see lines in the spectrum.
- will be the first line because the I told you we represent in centimeter inverse energy is increasing this way and that this will require the minimum energy.
- Nevertheless the intensity of this line is going to be very high.
- So you can see here a thick line over here because most of the molecules intensity always depends on the population of the state.
- And since requires minimum energy, though it is a short line it would give you an impression that this.
- Will be the weakest line, but no here the energy gap.
- It's short only because the energy gap is less.
- Therefore, and therefore it appears as the first line.
- And with low energy, and the second line is your first overtone, which is 0 to 2, and that line is appearing here.
- Have 0 to 3 and you have the turn line after this, it actually the intensity goes on falling down, and but one more observation which you have to make here that because these energy levels are coming closer and closer, we will find that the energy.
- Energy gap also decreases as we go further and further in the spacing between the energy lines is also becoming lesser and lesser.
- If you see the lines converging like this, it means that the molecule is showing anharmonic behavior.
- This is what we see at room temperature.
- V is equal to zero, the V is equal to one level also got populated because of the high temperatures.

- Some molecules gained energy and now they are at V is equal to one level.

Example or application from the lecture

- we are expressing energy in terms of centimeter inverse and that's what I told you in the first class that we use spectroscopic units it is just for convenience okay now let us calculate for first overtone so.
- Let us observe what happens.

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EV0000102 · EV0000103 · EV0000104 · +1 source units in pack.json

Core explanation · 20:00–24:59

And light infrared radiation is going to fall on this such molecule, then we will find that there will be some transitions happening from V is equal to one also.

Key details

- Minus 3, we will find lines originating from V is equal to 1 as well.
- Nevertheless, the intensity will still be not so high, it will still be less, and therefore.
- We will find that such lines will appear in the spectrum if we do it at a higher temperature, and the reason being the higher vibration levels also are getting populated, and rightly so they are called as hot bands because you see them at higher temperature.
- When you substitute in the energy expression where it changes from goes from 1 to 2, we get this expression nu e bar 1 minus 4 x centimeter inverse.
- If you remember fundamental absorption was nu E bar 1 minus 2 XE.
- This was for the fundamental absorption.
- Is going to this energy gap is further less.
- So, therefore, where will it appear in the spectrum?
- It will appear to the left of the fundamental absorption in the spectrum.
- In the next class, we are going to study about another type of spectroscopy, which is the electronic spectroscopy.

Example or application from the lecture

- Now, if we say let us take an example, we it is at V is equal to 1 and it is going to V is equal to 2.

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EV0000106 · EV0000107 · EV0000108 · +1 source units in pack.json

7 Electronic spectroscopy - Franck-Condon principle (Part II)

Study focus: Electronic spectroscopy - Franck-Condon principle (Part II). This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment.

EV0000110 · EV0000111

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module one on molecular spectroscopy.

Key details

- This will be the last class of this module.
- Today's class we are going to discuss about the intensity of spectral lines and the Frank Condon principle.
- We are studying electronic spectroscopy.
- Which means that when we allow UV visible light to fall on the molecule, the electron can undergo transition from the ground state to the excited state.
- And within the electronic state we know that there are vibrational level and these vibration levels you can see are.
- Coming closer and closer, the energy gap between the levels is coming, is becoming less, which shows that the molecule is an harmonic in nature.
- Also, the energy profile is a molecular.
- curve that shows us that the molecule is undergoing anharmonic oscillations and V double prime equal to 01234 is representing the vibrational levels in the ground state and v prime equal to 0123 represents the vibration levels in the excited state.
- It means that when an electron undergoes transition from the ground state to the excited state, it can end up in any of these vibrational levels.
- There is no rule forbidding any vibrational change.
- Vibrational levels of the excited state.
- So that means the spectrum is going to be very complicated.
- but we have a rule which helps us simplify the spectrum that most of the transitions start from V double prime equal to zero as it is the most populated level.
- And that becomes the most populated level.
- And therefore, most of the transitions will start from V double prime equal to zero.
- These vibration levels in the excited state, that's because of the rule one.
- Now the third rule says the excited state usually has longer internuclear distance because of the anti-bonding character.
- And if you look at the internuclear distance of the excited state, we see that it is definitely larger.
- So if the antibonding character is more, the number of antibonding electrons are more and therefore we get a small bond order.

- And we know that bond order and bond length are inversely proportional.
- So if the bond order is small, that means the bond length is going to be more.
- The next rule says the transitions arise from the center of V double prime equal to 0 because the maximum probability of finding the nuclei is at the center of V double prime equal to 0.
- They should be actually V double prime equal to zero, V double prime equal to one, two, three, four.
- But because we have shown only one electron state, we have written it as V is equal to zero, one, two, three.
- But since we are talking of electronic transitions.

Example or application from the lecture

- We know that bond order from molecular orbital theory we know that bond order is given by number of bonding electrons minus the number of antibonding electrons divided by two.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000110 · EV0000111 · EV0000112 · +1 source units in pack.json

Core explanation · 05:00–09:59

we should denote the ground state as V double prime equal to zero.

Key details

- The psi squared becomes the probability of finding the nuclear.
- For V is equal to 0, we find that this is the shape.
- And here we see that in the center there is maximum probability of finding the nuclei.
- As we go to V is equal to 1, we find that it becomes like this.
- And then as we go to higher and higher values of V, we find that the probability of finding the nuclei is more at the extremity as for V is equal to zero, the probability of finding the electrons in is more in the center.
- And therefore that is going to help us further understand the pattern.
- V double prime equal to zero is the one which is most populated level at room temperature and.
- Here from the probability graph we can understand that the center is where we are going to have the electrons.
- It is not that there are no electrons here or the nuclei do not exist in these positions at any point of time, but more probably of finding the nuclei is at the center for V double prime equal to zero.
- Rotational spectroscopy, vibrational spectroscopy, as well as electronic spectroscopy and the rules which are governing the energy profile and the placement of these.
- s energy levels.
- So for both ground as well as the excited state and therefore we know that it is.
- And harmonic in nature, and also the vibrational levels seem to be coming closer, and that also is an indication that we are talking of anharmonic anharmonically oscillating molecule.
- The ground state we have these vibrational levels, and within the excited state also we have these vibrational lepths.

- Within these vibrational levels we have the rotational levels.
- J value, then we find that these rotational levels become further and further apart.
- And that is what we have understood.
- Within each of these electronic levels we have the vibrational levels, and within the vibrational levels we have this rotational levels.
- If I want an electronic transition to take place, or if I want the molecule to move from the ground electronic state to the higelectronic state, it requires a lot of energy and that energy is corresponding to UV visible light.
- Whereas if I want just a vibrational change in the molecule, then this small amount of energy is what is required and that happens to be in the infrared region.
- why we draw the energy level diagram this way and also we know that because of the anti-bonding character there is a slight displacement in the excited state.

Example or application from the lecture

- If you remember in the initial classes I had given this diagram, but we did not clearly understand why we are drawing the diagram in this way for a diatomic molecule.
- Let us now look at Frank Cordon principle, which is going to explain the intensity of the spectrum.

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EV0000114 · EV0000115 · EV0000116 · +1 source units in pack.json

Core explanation · 10:00–14:59

So that means all lines should be of equal intensity, but it is not found to be so.

Key details

- That is explained with the help of Frank Contin principle, which states that an electronic transition takes place so rapidly that a vibrating molecule does not change its internuclear distance appreciably during the transition.
- the electron being very light undergoes transition very quickly and the nuclei being bulky they take time to adjust to this transition and therefore for the electron it is as if the nuclei are stationary.
- That in this case, what you see here that the electron has moved from V double prime equal to 0 to V prime equal to 2.
- And therefore, since.
- V double prime equal to zero is the state which is most populated and at the same time we know that transitions will happen from the center of V double prime equal to zero.
- So most of the transitions are happening from happening from here and therefore two zero line will be most intense in this.
- RE double prime and r e prime are the internuclear distances of the ground and excited states respectively.
- And RE double prime would correspond to the internuclear distance for the ground state whereas R prime would stand for the internuclear distance of the excited state.

- The first one states that the internuclear distance of the excited state is more than that of the ground state.
- Double prime equal to zero is the most populated state and transitions most of the transitions will happen from the center because that is the where we have the maximum probability of finding the nuclear.
- So therefore in this case we find that.
- Two zero is the line with maximum intensity, but that doesn't mean we don't have other lines, there are other lines also, but the probability of finding the nucleide electron is much less.
- in these areas and therefore we find that there are lines corresponding to 1000 and also three zero four zero etc but though those intensities are much less as compared to two zero and here we see that two zero line is here and this is centimeter inverse meaning that energy is increasing in this way one zero definitely would be to the left of two zero.
- Because the position represents the energy gap.
- And therefore we find that one zero is less than in energy so we will show the line here but nevertheless the intensity is less because the intensity is always more in the center of e double prime equal to 0 and similarly 0 will be to the left 30 though the intensity is less it is appearing to the right of 2 0.
- Another case would be when there is no change in the internuclear distance.
- In that case, again there is going to be a vertical transition and we find that.
- It happens from the center and ends up in the center of V prime equal to 0.
- And therefore the maximum intensity now will be for the 00 line.
- And there are other possibilities that the probability of finding the new.

Example or application from the lecture

- Let us study this in little more detail.
- Let us see what they are.

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EV0000118 · EV0000119 · EV0000120 · +1 source units in pack.json

Core explanation · 15:00–19:59

Does exist on either side and therefore we find one zero two zero three zero lines are also seen in the spectrum.

Key details

- Nevertheless, one zero will be to the right of zero because zero requires the minimum energy, one zero will require little more energy.
- And since the spectrum showing the energy scale in this direction, therefore we find that one zero will be here.
- then in that case zero nine is that it has the maximum intensity.
- The third case would be where the internuclear distance of the excited state is considerably greater than that of the ground state.
- So that means RE double prime is much less than RE prime.

- You can see that it is very much displaced to the right.
- Dissociation energy of the excited state.
- These atoms gain kinetic energy, and now the energy is not quantized anymore.
- This kinetic energy.
- Energy is not quantized, and when energy is not quantized, we cannot get discrete lines, and what we see is a continuum.
- You can see in the spectrum here, we see a continuum in the spectrum.
- Nevertheless, there are some lines to the left of the continuum.
- And those are of lesser intensity because there is probability of finding the nuclei around the center also.
- But nevertheless, the maximum intensity is from the la for the line which is originating from the center.
- But since gone beyond the dissociation energy, what we see is a continuum in the spectrum.
- That brings us to the end of module one.
- In the next class, we are going to talk about nanomaterials.

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EV0000122 · EV0000123 · EV0000124 · +1 source units in pack.json

8 Electronic spectroscopy(Progression)

Study focus: Electronic spectroscopy(Progression). This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000126 · EV0000127

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module one on molecular spectroscopy and we are going to study electronic spectroscopy in this class.

Key details

- We will revisit the Born-Oppenheimer approximation.
- And then finally we will study the vibrational core structure, which is also called progression.
- In electronic spectroscopy, we will find out where the electronic levels in the molecule.
- Like in rotational spectroscopy, we talked about rotational levels.
- In vibrational spectroscopy, we talked about the vibration levels.
- In electronic spectroscopy, we will talk about the electronic levels which are there inside the molecules.
- and now when a UV visible light is made to fall on the sample, the energy is sufficient to cause electronic transitions and these transitions will happen from the highest occupied molecular orbital to the lowest unoccupied molecular orbital.

- For these electronic transitions to take place, UV visible light has the correct frequency which matches with the frequency or the energy gap between these electronic levels.
- In molecular orbital theory we studied that we have sigma molecular orbitals, pi orbitals, then we have n and then we have the pi star and then the sigma star.
- And n is known as a non-bonding orbital.
- You can also have pi to pi star and you can have n to pi star.
- If the energy gap is very high, then we it will fall in the UV region and therefore we cannot see the color.
- But if these transitions happen to fall in the visible region or the energy gap is same as that of the visible region, then we will be able to see the color.
- of the particular molecule.
- There is electron redistribution, meaning if the electron goes in one of the molecular orbitals, which is the highest molecular or highest occupied molecular orbital, and there is a transition which happens to the lowest unoccupied molecular orbital, then this there is a electron redistribution.
- which is going to change the electric field.
- And therefore we find that all molecules can undergo electronic transitions.
- If you remember rotational and vibrational spectra were not possible and we could not record those spectra because for the homonuclear diatomic molecules but.
- Electronic transitions are possible for all molecules.
- Because now we understand what do we mean by rotational levels, what we mean by vibrational levels, and what is a electronic level.
- Each of these energies can be treated independently.
- You can calculate what is the rotational energy of the molecule, what is the vibrational energy of the molecule, and what is the electronic energy of the molecule, sum them up and that will give us the total energy of the molecule.
- Now, if we are talking about change in total energy, because what we are interested in is the energy gap.

Example or application from the lecture

- Let us revisit the Born Oppenheimer approximation.

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EV0000126 · EV0000127 · EV0000128 · +1 source units in pack.json

Core explanation · 05:00–09:59

And the electronic change in energy of the molecule if this is given in joules.

Key details

- But most of the times in we express energy in spectroscopic units, therefore we write it as small E, so del E total is equal to the del E rotational, del E vibrational plus the.
- Del E electronic and remember these are now expressed in wave numbers.
- That is because the energy gap between the electronic charge electronic levels is much higher than that of the vibrational levels.

- vibration into ten to r power of three.
- So that means the energy gap in an electron in the like between electronic levels is almost thousand times that b what is there between the vibrational levels and similarly for rotational levels it is e del e rotational multiplied by ten to the power of six.
- So this is the reason why the Born Oppenheimer approximation holds because what are we what is the approximation here that each of these energies are totally independent of each other because the difference in these energy levels is very, very high.
- Within the vibrational levels we have the rotation levels.
- You can see very clearly the difference in energy is very high.
- or and within the vibration levels and rotation levels.
- That is true.
- Why because they could be in the same electronic state, but they could be in different vibrational levels within the same electronic state.
- And also they could be in different rotational levels within the same vibrational state.
- So therefore the total energy would be the sum of.
- Which electronic state it is in and what is the vibrational state and what is its rotational state.
- is saying that an electron can ha can be in the same electronic state, but the total energy could be different.
- Now electronic transition as I mentioned just now it involves electronic redistribution means the electron is now it was at one particular level and it went to another level.
- Therefore, there will be a distortion in the electrostatic forces or in distortion in the electric field which is felt by the molecule.
- It is going to vibrate.
- Whenever there is an electronic transition, there is a vibrational change also happening, and there is a rotational change also happening.
- And if we have very good resolution, we can even see the rotational changes which are happening which is referred to as the fine structure in the electronic spectr of the molecule.

Example or application from the lecture

- So therefore we can use this approximation that the total energy of the molecule can be given as a sum independent sum of the rotational vibration and electronic energies and assuming that there is no electro there is no coupling between any of these energies.

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EV0000130 · EV0000131 · EV0000132 · +1 source units in pack.json

Core explanation · 10:00–14:59

But along with the electronic transition, there could be some vibrational changes.

Key details

- If we can somehow record those vibrational changes, we can get the core structure that's called the vibrational core structure and if the resolution of our instrument is very high we can even see the fine structure.

- Remember right in the first class I said that molecular spectra are band spectrum so now you can understand that the band is the electronic transition within that when you see these lines these fine lines they are due to the core structure and further if you see finer lines then that is referred to as the.
- So that means.
- For homonuclear diatomic molecules like hydrogen and nitrogen, which were otherwise inactive in the IR and microwave region, we can record the electronic spectrum and if we can get the vibrational and the rotational structure from this electronic spectrum.
- Then we will be able to get all the parameters, the rotational parameters, the vibrational parameters for these homonuclear diatomic molecules.
- pure rotational and pure vibrational spectroscopy but unfortunately the hydrogen and nitrogen molecules which are homonuclear diatomic molecules they are not active because they don't possess dipole moment and there is no change in dipole moment when the molecule either rotates or when it vibrates.
- But all these molecules they have these molecular orbitals filled with electrons.
- So electronic transition are possible and therefore using the electronic spectrum we can get to the vibrational core structure and the fine structure and that is.
- How we can determine all the rotational and vibrational parameters for these molecules.
- How are we going to use it to determine the vibrational parameters of the molecule, though?
- We are using UV visible light, that means it is going to result in electronic transitions.
- This is the structure where we have this lower electronic state that is known as the ground state, and this will be the excited state.
- And what we notice here that the lines are becoming closer and closer.
- What are these lines representing?
- They are representing the energy levels.
- Vibrational energy levels within the electronic state.
- What does it mean that.
- These are this is representing an anharmonic oscillator and we notice that V double prime equal to zero happens to be slightly away from the e is equal to zero because we know that always molecules will possess this zero point energy.
- prime we are referring to the ground state and V prime refers to the excited electronic state.
- So, therefore, we have V double prime equal to 01234, and here we have V prime equal to 01234.
- And just like we have vibrational levels in the ground state, we will also have vibrational levels in the excited state.
- And therefore, it is very important that we distinguish in the nomenclature.
- And when I say V double prime equal to one then it is referring to the ground state.

Example or application from the lecture

- The always the ground state is given the double prime means E double prime stands for the ground electronic state and e prime always stands for the excited state.

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EV0000134 · EV0000135 · EV0000136 · +1 source units in pack.json

Core explanation · 15:00–19:59

For vibrational transitions during an electronic transition.

Key details

- If whenever there is a molecule which is going from ground state to the excited state, we must remember that we are using UV visible light, that means the transition has to happen from here to here and not within the vibrational level, say or of the same.
- electronic state no we are using UV visible light so therefore the transition has to happen to the next electronic state now it can go from zero to zero it can go from zero to one zero to two and also it can go from V double prime equal to one and go to two three four of the higher levels.
- What does it mean?
- There are no selection rules to be followed.
- When we are talking about an electronic transition, during an electronic transition, there is no selection rule for the vibrational transition.
- But again, this fact helps us to simplify the spectrum.
- Because we know at room temperature it is V double prime equal to zero.
- Now we should not write V is equal to zero, we should say V double prime equal to zero because we are talking of two different electronic states.
- So V double prime equal to zero is the most populated state populated level and therefore all transition are going to start from v double prime equal to zero.
- you can have V double prime equal to 0 to V prime equal to 0.
- This is how the energy gap is increasing.
- All these are permitted because there is no selection rule.
- And also now all the transitions are starting from V double prime equal to zero.
- We just now saw that there will be transitions and how do we label these lines?
- It's very important to know how we are labeling them.
- comma V double prime what does it mean that the lines are going to be labeled according to the level where it goes that is this is the higher vibrational level vibration level of the high excited electronic state and this is the vibration level where it is originating.
- So therefore if there is a transition from zero to zero then we will get a line here that is zero.
- And if the transition is from zero to one.
- So it is one zero because we always call label the lines like this the higher.
- And again, we can see centimeter inverse is the scale here.
- That means the energy is going this way.
- I have to put the 00 line here first and not the 1, 0 or.
- The reverse way.
- Because the energy gap is going on increasing as we move from left to right.
- Because the numbers 0, 1, 2, 3, 4, they are changing in such a manner to where it is increasing by one for every line here.
- Understand all these lines are originating from zero.
- So therefore zero, one zero, two zero, three zero and four zero would rebuild the lines you will see in the spectrum and such a spectrum is known as a progression.

Example or application from the lecture

- Let us come to the next rule.
- Electronic state vibration level is named first, is it given first and then the lower.

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EV0000138 · EV0000139 · EV0000140 · +1 source units in pack.json

Core explanation · 20:00–24:59

fact which you see here as we go to these other hights where the energy is more we find that the lines are coming closer together.

Key details

- So that is because there is an harmonicity in the excited electronic state.
- If it was harmonic, then all these lines would be equidistant.
- But since there is an harmonicity in the upper electronic state, we find that these lines are actually converging.
- They will start coming closer and closer as we go to the higher values of energy.
- We have understood this very clearly.
- This is exactly what we drew and this is the ground state and this is the excited state.
- electronic state and there are transitions happening all our transitions are starting from v double prime equal to zero so if I say it is a line to zero that means it is starting from v double prime equal to zero and which and it is entering up at V prime equal to 2.
- So that would represent a two zero line similarly any of these lines are can be seen here these are all electronic transition and yes this requires minimum energy zero and then energy gap goes on increasing therefore the lines also will appear towards the higher energy.
- And then we have these lines coming closer together as we go to the higher values of energy, and this shows very clearly that the there is an harmonicity in the excited state.
- From this spectrum, we can get parameters.
- of the excited state very clearly.
- It appears for a very short while so therefore calculating the various parameters is going to be very difficult.
- But if you can record this electronic spectrum, then we can get all the information about the bond length, bond strength.
- If you're doing only the vibrational core structure, you can get the bond strength, and if you can do the rotational fine structure as.
- All lines for the ground state are originating from V double prime equal to zero, but nevertheless they are ending up at different levels of.
- The excited state and therefore we can get information about the excited state if we can record the spectrum.
- One is about this excited state, any transient.
- Excited state, we can get all the parameters, and for homonuclear diatomic molecules also, we can get all the information.
- In the next class, we are going to talk about the intensities of these lines.

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EV0000142 · EV0000143 · EV0000144 · +1 source units in pack.json

9 Phase, Component, Degrees of freedom, Statement of Gibb's phase rule, chemical potential

Study focus: Phase, Component, Degrees of freedom, Statement of Gibb's phase rule, chemical potential. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000146 · EV0000147

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will start with unit two.

Key details

- First, we will look at module three on phase equilibrium.
- The module content goes like this.
- We will understand what is phase equilibria.
- We will derive Gibbs phase rule.
- Using the Gibbs phase rule, we will construct the phase diagram of a one component system and then go on to apply the Gibbs phase rule to a two component system and construct the phase diagram.
- And then we'll study the iron carbon phase diagram.
- We will introduce a new term called chemical potential and then we look up certain definitions in phase equilibria that is about the phase component and degree of freedom.
- Free energy in phase equilibrium.
- What exactly do we mean by phase equilibrium?
- We have different phases of matter, you're aware, of solid, liquid, and gas.
- And we find that in certain conditions the solid phase is more stable.
- In other conditions, liquid phase is more stable, and in some other conditions, gaseous phase becomes more stable.
- We know that below zero degrees we find solid phase ice is stable.
- At from 0 to 100, we find it's the liquid phase, and then beyond 100, it is the water vapor.
- What is the driving force?
- Why is it that below zero degrees we find only the solid phase and not the liquid phase?
- Whichever phase has the lowest free energy, that phase is going to be present.
- We know that all systems try to reduce their free energy and.
- In this case, we find that solid is having lower free energy than liquid at zero degrees.
- When you cross zero and up to 100 degrees, we find that the liquid phase has minimum free energy, and similarly, gaseous phase will have less lower free energy when it comes to beyond

100 degrees.
- So every system tries to get minimum free energy, and therefore the phase which has minimum free energy will be most stable phase.
- That is, solid changes into liquid.
- Ice is getting transformed into the liquid phase.
- beyond the melting point.
- When you heat the system beyond the melting point, it the solid changes into liquid.
- Why would such a transformation happen?
- Between 0 to 100, it is a liquid phase which is more stable.
- And what would that stable phase mean?
- Just now I said it will have a lower free energy.
- which has a higher free energy to one which has lower free energy.
- Reduce free energy of the system.
- That means Gibbs free energy plays a very important role in deciding what phase transformations will happen under a certain set of conditions.
- But Gibbs energy is an extensive property.
- What is chemical potential?
- It's basically gives free energy divided by the number of modes.
- The moment you do that, you have made it into an intensive property, and now you can compare systems, phases, etc, with the help of this new term which is known as.
- So just saying gives free energy divided by the number of moles that is molar Gibbs free energy would not be sufficient because you have so many components, we say chemical potential, we define it as partial molar Gibbs free energy.
- It is not just Gibbs free energy, it is partial molar Gibbs free energy.

Example or application from the lecture

- The most common example we can take is that of water.

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EV0000146 · EV0000147 · EV0000148 · +1 source units in pack.json

Core explanation · 05:00–09:59

How would I mathematically give an expression for this partial molar gives free energy.

Key details

- And remember it is denoted by a symbol mu.
- So for this component chemical potential is defined as del G by del Ni.
- NI is the number of moles of that particular component in the system.
- And since a partial derivative, so pressure, temperature, and number of moles of other components are taken as constant.
- Whenever we say phases are in equilibrium, now instead of Gibbs free energy, we'll refer to chemical potential.
- So, what would be the condition for phases to be in equilibrium?

- The chemical potential of a particular component should be the same in all the phases which are present in equilibrium.
- So that means at the triple point of water where you have solid, liquid, and the vapor in equilibrium,.
- What is the condition for phase equilibrium?
- The chemical potential of the solid form is equal to the chemical potential of the liquid form, which is equal to the chemical potential of the vapor form.
- what would happen in that case, then at some point in some phase the chemical potential is more as compared to the chemical potential in another phase.
- So what would happen because of that?
- There would be a transformation, phase transformation from higher chemical potential to lower chemical potential.
- If it is a heterogeneous system, of course, when we say different phases are present, it is a heterogeneous system.
- There can be no transfer of energy or mass from one phase to another.
- Only when there is no transfer of mass or energy, we say the system is at equilibrium, or.
- Another way of looking at it is when a system is in equilibrium, the various phases must have the same temperature and pressure, and the composition must remain constant all along.
- These are the different ways by which we can understand phase equilibrium.
- A phase is defined as a homogeneous and physically distinct part of a system.
- And there will always be a boundary between phases.
- Whenever you see a boundary, when one phase from the other, there will be a boundary.
- And it is denoted by P, and P is a number.
- Whenever we say phase, we are referring to the number of phases in a particular system.
- Now, if we have a liquid system, the number of layers, we know that when liquids are immissible, the layers going to be seen in that system, and therefore the number of layers will decide the number of phases.
- Provided they are invisible.
- When we have a solid state, then each state, each of these solid will constitute a separate.
- Phase we have to count each solid as a separate phase.
- All solids are not taken as one phase because how did we define phase?
- There is a boundary.
- They are mechanically separable.
- Therefore, each solid is going to constitute one phase.
- And not only that, each polymorphic form or allotropic form.

Example or application from the lecture

- Three phases, then in that case we find that suppose it is solid, liquid, and vapor, then we find what is the requirement for a system to be at equilibrium, that the chemical potential the chemical potential of a component is equal in all the phases which are present in equilibrium.
- First, let us take up phase.

- If they are completely miscible, for example, alcohol and water, then P is going to be equal to one.

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EV0000150 · EV0000151 · EV0000152 · +1 source units in pack.json

Core explanation · 10:00–14:59

Is also going to constitute a separate phase.

Key details

- What do we mean?
- Carbon exists as graphite as well as diamond.
- Both of them cannot be treated as one phase, or we cannot say p is equal to one here.
- Because both of them have boundaries, they can be easily separated.
- And if there is a solid solution as in isomorphic alloys, then P is taken as one.
- Here we see P is equal to C three.
- Because it may be the same system.
- It may be.
- It may be water system.
- Remember that two solids will contribute to two different phases and one gas.
- So we have two solid phases, calcium carbonate, calcium oxide, and P is equal to three now because there's also carbon dioxide gas.
- In this case, again, P is equal to 2.
- Because solid is one, but gases are completely miscible, therefore ammonia and Cl together will constitute one phase.
- A component is defined as the smallest number.
- We need to pay attention to these.
- parts which have been underlined, smallest number of independently varying chemical constituents using which the composition of each and every phase.
- These two are to be marked very carefully.
- Smallest number and each and every phase of the system we should be able to describe with the help of these components.
- This is a number and is denoted.
- C will be the number of components.
- And when there is no reaction taking place, obviously as many number of constituents will be equal to the number of components.
- And since a number, one important.
- Fact which we have to remember is that when we are expressing in terms of constituent zero and negative values are also allowed.
- Water is a one component system where C is equal to one.
- Even if say we have two phases in equilibrium, that is at the melting point, the composition of both the phases will be the same.
- If since it's a single constituent, there is both of them will have the same composition.
- Both of them will be the same.

- We may have what are the phases here?
- What are the number of phases?
- We have two phases here.
- Solid water as well as liquid water, but both can be expressed in terms of a single constituent.
- That is a salt hydration.
- Salt hydrate system.
- Salt hydrate meaning what?
- It has a water of hydration.
- Because NA2SO410H2O can be expressed as the sum of NA2SO4 plus 10H2O.
- Only Na2SO4, if, that is not sufficient for me to describe this phase.
- Another phase of the same salt hydrate could be Na2SO47H2.
- We will write it as Na2SO4 plus 7H2.

Example or application from the lecture

- For example, we have carbon.
- Like this is given by this equation, what is the number of phases?
- Now let's come to the next definition, and that is of a component.

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EV0000154 · EV0000155 · EV0000156 · +1 source units in pack.json

Core explanation · 15:00–19:59

If we take the anhydrous salt, then what will happen?

Key details

- NE2SO4 is NE2SO4 plus 0H2O.
- As I mentioned, in the case of components, 0 can also be used.
- We are calling it a two-component system.
- One component is enough to describe this particular phase, but we define component as what?
- The total number of phases, the minimum number, the smallest number which is described which is used to describe all the phases.
- You should be able to describe all the phases using that minimum number.
- Here, though it is one, minimum you need is two to describe.
- All the phases.
- Evaporate the water completely and collect it, then you will have only the water phase.
- Meaning how does or how do all these phases come about here?
- Because you are changing the conditions, you can have different forms of the same salt hydrate.
- And how do we describe these phases?
- In order to describe the phases, now when you come to water, you will say zero of any two so four and water.
- So that means you need at least two components to describe this particular system.

- Again here we say C is equal to 2.
- Because though we have three different phases or three different constituents, there's an equation which is binding them together.
- We need not use all the three constituents.
- Two components are sufficient to describe all the phases.
- What are the phases?
- We have calcium carbonate solid, calcium oxide solid, and CO2 gas.
- Then the first phase CACO3 solid will be described as CAO plus CO2.
- CAO itself will be described as CAO plus 0CO2 and CO2 can be described as CAO plus CO2.
- How will I describe the phases now?
- We have CACO3 and CO2.
- But if I have to describe CAO now, I don't have CAO with me.
- How will I use these two?
- I can say CACO3 minus of CO2.
- In this case two components are sufficient for me to describe all the three phases.
- Here, though the reaction looks similar to the calcium carbonate reaction, C becomes equal to 1.
- What are the phases?
- Here we have only two phases.
- Remember, the one is a solid phase of ammonium chloride, and the other is the gaseous phase where both ammonia and HCl are present in the molar ratio 1 is to 1.
- Mixture of NH3 plus H Cl, but the num the phase is only one because gases are completely miscible.
- In this case, the solid phase is ammonium chloride.
- But when we come to the gaseous phase, we have ammonia and HCl in equal quantities.
- So therefore, ammonia.
- A plus HCl can also be represented as ammonium chloride.
- That's so in this case one component is sufficient for me to describe okay.
- I could use two components also.
- What ammonia and HCl.
- If I want to use ammonia and HCl, it will become ammonia plus HCl in this case, and the gaseous form will also be ammonia plus HCl.
- But the by definition component is the smallest number of constituents which are required.

Example or application from the lecture

- Let us see.
- Let us see if we can describe all the three phases using any two constituents.
- Let us take up another case where we have thermal decomposition of ammonium chloride in a closed system.

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EV0000158 · EV0000159 · EV0000160 · +1 source units in pack.json

Core explanation · 20:00–24:59

The composition of both these phases can be described with C is equal to one.

Key details

- But this is if it was a closed system.
- You have added certain extra amount of HCl or ammonia, so that has to be accounted for.
- Therefore, C becomes equal to 2 in this case.
- Just ammonium chloride is enough for the solid phase, but not enough for the gaseo phase because we added an extra amount of HCl or ammonia.
- The degrees of freedom or variance of a system is defined as the minimum number of intensive variables, such as temperature, pressure, or concentration, that must be ascertained in order to describe the system.
- In other words, we can also say what is my degree of freedom to change these variables independently without altering the number of phases, or rather without altering the equilibrium in the system.
- At a particular time, there is only one phase, so solid, liquid, or the gas.
- the without affecting the system, I can vary the temperature or pressure.
- In fact, solid, liquid, or gas form of water can be present at different temperature and pressure.
- Therefore, the degree of freedom or the variance is F that is equal to 2.
- Two phases in equilibrium where now temperature or pressure can be.
- They read or can be specified.
- What does it what do we mean?
- The degree of freedom now becomes one.
- When does this happen?
- This happens at zero degrees centigrade, but the moment I say zero degrees centigrade, the pressure becomes fixed to one atmosphere.
- If I change the pressure, then the melting point is also going to change.
- That zero degrees will not remain zero degrees.
- Though we have solid water and liquid water in equilibrium, now I can either change the temperature or I can change the pressure.
- Therefore, when two phases are in equilibrium, f becomes equal to one.
- Here, none of the variables can be changed, meaning that neither the temperature nor the pressure can be changed, and therefore we say f is equal to.
- That would be a triple point of water.
- Solid water is in equilibrium with liquid water, which is in equilibrium with water vapor.
- All the three phases are present together, they coexist in equilibrium, and now.
- Neither the temperature nor the pressure can be changed because those are fixed quantities.
- And what is that value?
- 0.0098 degrees centigrade and 4.6 millimeter of mercury for water.
- In a system, F becomes equal to 2.
- You can change temperature and pressure independent of each other.
- When you come to two phases in equilibrium, in that case, you can either change the temperature or you can change the pressure so f is equal to one and when you have three

phases in equilibrium then f becomes equal to zero.
- You can neither change the temperature nor you can change the pressure.
- Remember they are numbers.
- So now in the next class we are going to look at.
- Gibbs phase rule, which is going to combine all these three components, phase and degree of freedom, and give you a very nice elegant thermodynamic equation.

Example or application from the lecture

- Let us look at how to understand this.
- Let us take up the water system.
- Suppose we have two phases in equilibrium.

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EV0000162 · EV0000163 · EV0000164 · +1 source units in pack.json

10 Derivation of Gibb's phase rule, Phase diagram of one-component system - water - Part II

Study focus: Derivation of Gibb's phase rule, Phase diagram of one-component system - water - Part II. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000166 · EV0000167

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module three on phase equilibria and today's class we will talk about phase diagram of water and we will discuss reduced phase rule for a two component system.

Key details

- And for a one component system, F is equal to 3 minus P.
- If we have two phases in equilibrium, then f becomes equal to one and that would be represented by a line and when.
- There are three phases in equilibrium.
- F could be equal to 0 and that would be represented by a point.
- And what do we have here?
- We have the solid in equilibrium with.
- Liquid and since there are two phases in equilibrium, this will be represented by a line, and then OA similarly would be the vaporization curve.
- What would that be?
- Water is in equilibrium, liquid water is in equilibrium with the water vapor.
- And OB stands for the sublimation curve.

- where we have the solidized in equilibrium.
- with the water vapor.
- And an important point which we should note here is the triple point O, which is where all the three phases are in equilibrium, the solid, liquid as well as the water vapor.
- And then there's another point here A, which is known as.
- As the critical point.
- We will just discuss this in the next few slides.
- And then we also have an equilibrium OA prime, which is shown by the dotted lines over here, and this is known as the metastable equilibrium, which we will discuss soon.
- O C in this diagram, which is also known as the melting point curve.
- It's a one component system and there are two phases in equilibrium.
- O F becomes equal to 1 and the system is a monovariant system.
- What does it mean?
- That between pressure and temperature we can change only one variable.
- If we change the pressure, the temperature is constant.
- Is defined, or if we change the temperature, then the pressure is defined.
- What does this line represent here?
- The variation of melting point of ice with pressure.
- As we change.
- The pressure as we increase the pressure, we find that the melting point of ice decreases and it's very clear that there is a negative slow.
- Most of the materials we find that the fusion curve or the melting point.
- Curve has a positive slope.
- But in case of water, the slope is negative because of the anomalous behavior of water.
- Therefore, we know that ice floats on water because the density of ice is less than that of water.
- All other substances, the solid is always has a higher density as compared to the liquid.
- Why did I choose dp by dt?
- Because here pressure versus temperature is what we have plotted and dp by dt would give me the slope of the.
- Go now in the case where we have ice is equilibrium with water, ice is changing into water.
- We know that we have to use delta H fusion or delta H melting.

Example or application from the lecture

- We have solid ice and liquid or water in equilibrium that is given by this line OC.
- Let us first discuss the equilibrium between solid and liquid, that is the fusion curve.

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EV0000166 · EV0000167 · EV0000168 · +1 source units in pack.json

Core explanation · 05:00–09:59

That heat is going to be absorbed, so therefore it is going to be an endothermic reaction.

Key details

- In the case of ice changing into water, because we know that there is going to be a decrease in volume as ice melts.
- Is negative in the case of water.
- One more important point we need to notice here that when ice changes into water, there is the value of delta V is going to be a very small value and therefore we find that the slope delta h by t delta v that is dp by dt has a very high value.
- Equilibrium between ice and water.
- Here the slope is going to be negative and the slope is going to be very steep.
- Now coming to the next equilibrium that is the solid and the vapor which is known as the sublimation curve here the curve OB.
- Here we have ice in equilibrium with water vapor and similarly when we apply the phase rule F is equal to one, so it's again a monovariant system.
- Here, what do we see along this?
- We understand the variation of the sublimation temperature of ice with pressure.
- As we change the pressure, what happens to the sublimation temperature?
- And here we find that the slope is positive.
- In the case of.
- We found that the slope is negative, whereas here we find the slope is going to be positive.
- Again using the Clausius Clabron equation we will be able to explain this.
- DB by D T is equal to delta H sublimation by T delta V.
- Now again delta H sublimation is going to be a positive value because when ice changes into water vapor, there is the reaction is going to be endothermic, it's going to absorb heat.
- And delta V is also going to be positive in this case when ice changes to water vapor because there is going to be a big change in the volume.
- When ice changes to water vapor, there's a large change in the volume.
- Both being positive delta H sublimation as well as delta V is positive, and therefore dB by Dt is coming up is going to be positive.
- And one more point to notice here is that delta V being large.
- This delta V is very large, therefore dp by dt slope is much less as compared to the melting point curve.
- Vaporization curve the equilibrium between the liquid and the vapor which is represented by this line over here.
- Again we have liquid water in equilibrium with vapor and f is equal to one, so it's a monovariant system, therefore it's represented by a line.
- We can either change the temperature or the.
- The variate variation of boiling point of water with pressure.
- As we increase the pressure, we notice that the boiling point increases.
- And therefore we see that the slope is positive using the.
- Closer's Clapebron equation again we can explain that dp by dt is equal to delta H vaporization by T delta V.
- Now delta H vaporization is going to be again positive because it's an endothermic reaction, liquid water changing into water vapor requires heat.
- And then T delta V, so delta V.

- It is again positive here, liquid water changing into vapor, again there is a big change or a large change in the volume, and therefore we'll find that delta V is also positive.
- And since delta V again is a large value we find that the slope is much less as compared to that of the melting point curve.
- Since all the three phases are present together, P is equal to 3.
- So therefore F is equal to which is equal to C minus P plus 2.

Example or application from the lecture

- Let us talk about the triple point O.

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EV0000170 · EV0000171 · EV0000172 · +1 source units in pack.json

Core explanation · 10:00–14:59

equal to zero, so the system becomes invariant.

Key details

- What does it mean?
- That we can neither change the temperature nor the pressure.
- Another point of importance is A, which is known as the critical point.
- At the point A the interface between the liquid water and the water vapor vanishes.
- That means we cannot distinguish between the liquid water and the water vapor.
- Such a point is known as the critical point.
- Of course this plot is not made to scale because we are using different pressure units.
- In fact we have even used different units for pressure.
- When water is cooled below the triple point, it is supposed to get converted into ice.
- But under certain conditions, it is found that water continues to be in the liquid phase and does not get converted into ice.
- This is known as super cooled water and it represents a metastable equilibrium.
- Which involves the liquid and the vapor phases.
- It is called metastable equilibrium because even a small disturbance is going to cause the system to go back to the stable equilibrium which is represented by the line OB.
- The vapor pressure of the system now in the metastable region is higher than that of the stable system of ice at the same temperature.
- And therefore, we find that the line OA prime is always shown above that of the sublimation curve.
- We know that the phase rule is F is equal to C minus P plus 2.
- But in the case of a two component system, we will not be able to use this ordinary phase rule.
- The reason being that.
- Since C is equal to two here, F is going to be four minus P.
- One phase then F becomes equal to three.
- So that means there are three degrees of freedom.

- Most of the time it is assumed that pressure is constant because we carry out measurements at atmospheric pressure.
- The new rule, therefore, is F is equal to C minus P plus one.
- And this is known as the reduced phase rule.
- The name is derived from the fact that we have reduced the degree of freedom by one because we are assuming.
- That one of the variables is constant.
- And this is mostly applicable to solid liquid equilibria because in this case the pressure changes.
- Will not affect the equilibrium to a great extent as compared to when there would be a gas phase present in the system.

Example or application from the lecture

- Let us discuss the metastable equilibrium represented by the line OA prime on this graph.
- Let us talk about the phase diagram of a two component system.
- We consider one of the three variables to be constant.

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EV0000174 · EV0000175 · EV0000176 · +1 source units in pack.json

Core explanation · 15:00–19:59

the condensed phase rule.

Key details

- So in the next class we are going to study the construction of the phase diagram for a two component system and specifically we will look at the lead silver system.

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EV0000178 · EV0000179 · EV0000180 · +1 source units in pack.json

11 Derivation of Gibb's phase rule, Phase diagram of one-component system - water - Part I

Study focus: Derivation of Gibb's phase rule, Phase diagram of one-component system - water - Part I. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000182 · EV0000183

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module three on phase equilibria and in today's class we are going to talk about Gibbs phase rule.

Key details

- We will derive the Gibbs phle and we will discuss phase diagrams specifically phase diagram of a one component system.
- If you remember, P stands for the number of phases, C stands for the number of components, and F stands for the degrees of freedom.
- The Gibbs phase rule is going to give us a relation between the number of phases, the number of components, and the number of degrees of freedom for a system existing at equilibrium.
- F is equal to C minus P plus 2.
- What are the conditions a system needs to satisfy in order to be considered to be at equilibrium?
- Firstly, it should be at thermal equilibrium, meaning that the temperature should be constant throughout the system.
- If there exists a difference in temperature, then.
- The system will try to equalize the temperature and it will reach equilibrium only when the temperature is same throughout the system.
- Talked about a new term chemical potential.
- When we talk of any chemical reaction, we know that gibbs free energy decides.
- the direction of the reaction.
- Wherever there is decrease in free energy, we know that the reaction is going to move in that direction because every system tries to go towards minimum free energy.
- But free energy cannot be used as a criteria for chemical equilibrium because it is.
- An extensive property.
- Therefore, a new term was defined which is known as the chemical potential.
- So for a system to be at chemical or material equilibrium, the condition is that chemical potential of the substance should be same in all the phases.
- Alpha is equal to mu i beta is equal to mu i gamma.
- What does it mean?
- That the chemical potential of the I component in the alpha phase is equal to the chemical potential of the I component in the beta phase, which is again equal to chemical potential of the I component in gamma.
- phase and so on.
- Our system is where all the C components are distributed between P phases.
- If we want to talk about any system at equilibrium, first and foremost, we should know what are the variables we need to ascertain.

Example or application from the lecture

- The phase rule was given by Williams Gibbs in eighteen seventy four.
- Before deriving the Gibbs phase rule, let us understand what we mean by a system at equilibrium.

- molar gives free energy which is chemical potential and since there are so many species in the system we have to consider partial molar gives free energy.

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EV0000182 · EV0000183 · EV0000184 · +1 source units in pack.json

Core explanation · 05:00–09:59

Firstly we will have temperature.

Key details

- We should know the temperature, so that should be one variable and then we should know the pressure, so that is again another variable.
- To know and also the pressure of the system which we need to know.
- The composition we need to know the mole fraction of each component in every face.
- For each phase, we know that the sum of the mole fractions is equal to unity, because this is a relative way of expressing composition, mole fraction.
- for all the components in that particular phase.
- Say alpha phase.
- In the alpha phase, we have so many components.
- There are C components.
- Plus mole fraction of the third component of the alpha phase, so on, till we come to the last component that is the mole fraction of the component C in the alpha phase.
- And all this should sum up to one.
- We are interested in the number of components.
- Why C minus 1?
- Because we know that all these will sum up to 1.
- Now coming to the second phase, say that is the beta phase, again we have an equation like this where we have the first component mode fraction, the second component mode fraction, third component mode fraction, so on adding up to one.
- And similarly for the gamma phase, and as we go on, we know that there are p number of phases.
- So therefore for the last phase it will be the mole fraction of the first component in p phase plus the mole fraction of the second component in p phase, so on.
- Up to the last component in the P phase, again this all adds up to one.
- Therefore, we have C minus one compositions again need to be determined.
- For each of the phases, we need C minus 1 more fraction terms which have to be define and since we have p number of phases therefore the number of composition variables will be p times c minus one.
- Plus we have temperature and pressure which we need to ascertain.
- So therefore the total number of intensive variables will become P times C minus one plus two.
- If we want to find out what are these variables.
- We need to know the number of equations or the constraints.

- At equilibrium, we know that chemical potential of a particular component is the same in every phase of the system.
- Just now we define chemical equilibrium and talked about this equation.
- So the first component chemical potential in the alpha phase should be equal to.
- The chemical potential of the first component in the beta phase and also that should be equal to chemical potential of the same component in the gamma phase and so on.
- So the first component chemical potential should be equal in all the p phases.

Example or application from the lecture

- So first let us talk of.
- So the equation would be given by the mole fraction of the first component in the alpha phase plus the mole fraction of the second component in the alpha phase.
- So let us look at it for the first component.

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EV0000186 · EV0000187 · EV0000188 · +1 source units in pack.json

Core explanation · 10:00–14:59

Second component chemical potential in alpha phase should be equal to the chemical potential of the second component in the beta phase, and also that again equal to the chemical potential of the second component in gamma phase.

Key details

- P because P is the total number of phases, therefore what we get here is again P minus one number of equations.
- Therefore, again we have P minus one number of equations.
- So therefore the total number of equints now we have is C times P minus 1.
- Relating X and Y.
- I cannot change X and Y independently.
- If X changes then Y is fixed if Y changes then X is fixed.
- Because of this one equation the degree of freedom or the.
- My independence to change X and Y is reduced by 1.
- And this is Gibbs phase root.
- We will take up the water system because that is the easiest for us to relate to.
- In that case, C is one because it is a one component system.
- P is one again.
- because we are talking about one phase and when we apply the Gibbs phase rule F is equal to two.
- That temperature and pressure can be varied independently.
- And when we have two phases present in.
- So either the solid is in equilibrium with the liquid or the liquid is in equilibrium with the vapor or the solid is in equilibrium with the vapor, then we know that there are two phases, C is again one because it's a one component system.

- So that means now the system has become a univariant system, and we know that when we have two phases in equilibrium, either the temperature or the pressure can be varied independently, but not both.
- And when we have all the three phases in equilibrium.
- And this system now, what happens?
- F is equal to 0.
- C is equal to 1, P is equal to 3.

Example or application from the lecture

- Since we know the number of variables and the number of equations, the degrees of freedom will be given by the total number of variables minus the number of equations which we have.
- Let us understand this.
- Determine now since there exists no relation, suppose there are independent variables, then that means I need to ascertain both X as well as Y in order to define the system.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000190 · EV0000191 · EV0000192 · +1 source units in pack.json

Core explanation · 15:00–19:59

an invariant system.

Key details

- It is a diagram which represents the conditions under which a substance exists in different phases in a system.
- How do I construct a phase diagram of a one component system?
- The phase rule we know which we are going to apply is F is equal to C minus P plus 2.
- If we have only a single phase, we want to represent a single phase in the diagram, we know that using this f is equal to 3 minus p, since it's one phase, f becomes equal to 2.
- And in a diagram, this would be represented by an area.
- When we have three phases in equilibrium then it will be represented by a point, but if you have just a single phase that would be shown by an area in the diagram.
- This area would be the solid phase.
- And in between we have the liquid phase.
- Say this area is going to be the solid phase, this area will be the gaseous phase, and in between we have the liquid phase.
- That would be represented by an area.
- Equilibrium between solid and the liquid.
- And similarly, liquid and vapor, so that would be this line here.
- And then if there is vapor and solid, then this line would represent the.
- And then if we have the solid liquid as well as the vapor present in equilibrium together, then that would be represented by a point, and that's exactly what we see here.

Example or application from the lecture

- Let us now discuss the phase diagram.

- Let us see what the phase diagram is going to look like.
- Suppose we have just one phase, and just now we said that when we have one phase, phase rule tell us that there is going to it is going to be represented by an area on the diagram.

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EV0000194 · EV0000195 · EV0000196 · +1 source units in pack.json

Core explanation · 20:00–24:59

as well as the vapor in equilibrium and that would be a point on the phram.

Key details

- So in the next class.
- We will take up the water system and see exactly what the phram is going to look like for a water system.

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EV0000198 · EV0000199 · EV0000200 · +1 source units in pack.json

12 Reduced phase rule, phase diagram of two-component system - Pb-Ag system

Study focus: Reduced phase rule, phase diagram of two-component system - Pb-Ag system. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000202 · EV0000203

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module three on phase equilibria and in today's class we are going to learn how to construct a phase diagram of a two component system in general and specifically the lead silver system.

Key details

- Then we study the determination of solid liquid equilibria and finally the Patterson's process.
- When we talk of a two component system, we know that we need to use the phase rule F is equal to C minus P plus 1, which is known as the reduced phase rule.
- It is known as the reduced phase rule because we have reduced the degree of freedom by one.
- Is equal to one, then the degree of freedom becomes equal to three, and we need to draw a three dimensional phram.
- Most often we use pressure as constant variable.

- So therefore because we reduce the degree of freedom by one, the reduced phase rule is now applicable.
- F is equal to c minus p plus one.
- And now out of pressure, temperature and composition, since pressure is being maintained constant, and especially for solid liquid equilibria, pressure is chosen constant variable.
- When we are drawing this phase diagram of a two component system, this particular phase diagram will be known as a simple eutectic system.
- One temperature scale stands for pure A and the other temperature scale stands for pure B.
- And on the x axis we have the composition scale.
- What is happening is here we have 100% A and as we move along this axis we are adding B into A.
- of various compositions which are represented by A and B, the percentage of A and B.
- Now the first thing we need to know is the melting point of pure A.
- Now that melting point because this axis represents pure A and the temperature scale.
- the alphabet M.
- Now as we go on adding B into A, the we are actually adding an impurity into the pure substance A, and therefore we know that there is going to be a depression in freezing point.
- When we add a certain amount of B, say a small amount of B, we find that it reduces the freezing point of pure A.
- And as we go on adding more and more of A, we find that this goes on decreasing like this.
- So therefore if I have to join these points I get a plot like this.
- And now if I add A into the pure B again I will notice that there is a decrease in the melting point and if I join these again I get a line like this.
- If I join these two they happen to meet at a point which is known as the utectic point.

Example or application from the lecture

- And for that, the criteria is that suppose we have two components A and B, then A and B must be miscible in all proportions in the molten state or the liquid state and in the solid state they are completely immissible.

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EV0000202 · EV0000203 · EV0000204 · +1 source units in pack.json

Core explanation · 05:00–09:59

If we draw a phram, then this is what the phram is going to look like.

Key details

- For that, what we do is the simplest way is to take any composition of A and B, mixture of A and V, and we heat it.
- beyond this point.
- So that means if I go beyond this point, then the liquid is what we are going to get is the liquid melt.
- So therefore we have the liquid melt now.
- What I do now is I start cooling it.

- When I come to this point here,.
- I find that we have that I have reached the melting point or the freezing point of A in the mixture.
- The first crystal of A will start forming here.
- Remember that this is not pure A, it is a mixture, and in a mixture now the melting point has reduced or the freezing point has reduced.
- As we go further down.
- We find that more and more of this.
- A is going to freeze out of the mixture.
- That's exactly what happens here, and how do we know what is the.
- Liquid melt now it is at this point say if.
- See, it has become richer in B.
- So as we go down further, we go on cooling it, more and more of A will precipitate out and therefore.
- Liquid melt has now reached this composition.
- When we reach this point of the graph, what we notice is that the liquid melt has now reached this point.
- We have reached even the freezing point of B.
- Therefore, we will find that B also starts precipitating at this point.
- The liquid is cooling initially and then we get the first crystal of A.
- As we cool down further, more and more A separates out and the liquid melt becomes richer and richer in B.

Example or application from the lecture

- More of A has come out, we find that the liquid melt can be given by this point.
- Let us now label each of these areas.

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EV0000206 · EV0000207 · EV0000208 · +1 source units in pack.json

Core explanation · 10:00–14:59

And both of them will meet at the same utactic point E.

Key details

- Because all these points represent the melting points of the mixtures.
- Therefore, this whole area will be the liquid melt.
- We find here that we have the liquidment as well as the solid A because in this area what is happening?
- Solid A is being frozen out and precipitating out and we have the liquid melt.
- So this area would represent liquid melt plus the solid A and similarly this area would represent the liquid melt plus solid B because if you are cooling on this side we will find that B starts respirating out.
- Liquid mill which again is has both A and B and here we have the solid B in this area plus the liquid mill.

- F E G.
- We have both solid A as well as solid B because below this point what do we observe?
- The liquid melt now completely converted into the solids A and B and here we see there is a.
- Apart from this, the another point of interest which I have been mentioning is the point E which is known as the eutectic point and also these curves are of interest to us the curve ME and the curve NE.
- found to be a eutectic mixture.
- It's a mixture of the two components A and B, which has the lowest freezing point of all the possible mixtures of the components.
- Here the name eutectic is from a Greek word which means easily melting.
- And since has the lowest melting point amongst all mixtures of A and B, this is referred to as the eutectic mixture.
- One is the liquid phase, one is solid A, and the other is solid B.
- Freedom at this point.
- Therefore F comes out to be zero.
- So therefore this point is invariant.
- Now we know that when we have an invariant point, it means that we cannot change the temperature or the composition.
- And one more important point which we need to understand here is that the melting point is very sharp.
- unlike other mixtures where the melting point is over a range of temperatures, in the case of a eutectic mixture it is found that it has a very sharp melting point.
- In fact, earlier it was assumed that it is actually a compound, because compounds usually have a characteristic melting point, but later on microscopic examinations and other evidence showed that this.
- Temperature, this composition is actually that of a mixture.
- It is not a compound, it is a mixture, hence the name eutectic mixture, but it has a very sharp melting point.

Example or application from the lecture

- Let us first talk about the eutectic point which is shown by the point E here on the graph.

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EV0000210 · EV0000211 · EV0000212 · +1 source units in pack.json

Core explanation · 15:00–19:59

The important features of the utected point is that the mixture has the lowest melting point amongst all other compositions, then the it is an invariant point and also it has a very sharp melting point.

Key details

- The melting point of pure lead is 961 degrees centigrade that would be represented here, and that of pure lead is 327 degrees centigrade.

- And the eutectic temperature and composition become fixed and the eutectic temperature is found to be.
- To be 303 degrees centigrade, and the eutectic composition is 97.4% of lead and 2.6% of silver.
- Now this ME curve first is known as the freezing point curve of A.
- I have already explained that as we go on adding more and more of B, the melting point or the freezing point of A is decreasing.
- A and it also shows that as we go on adding B into A, the freezing point or the melting point is decreasing.
- Here we have solid A which is in equilibrium with the liquid melt which has which has B in A.
- And now the number of components are 2, the degree the number of phases is also 2, therefore the reduced phase rule is applicable.
- F is equal to C minus P plus 1 is equal to so 2 minus 2 plus 1 and therefore now it is 1.
- F is equal to 1 along this curve and therefore it is a univariant system.
- A into B again here we have on the curve solid B in equilibrium with the liquid melt which is now A added to B.
- And again we will use the reduced phase rule we find that.
- F is equal to 1, so the system is univariant along this curve.
- Thermal analysis is the study of what we call as the cooling curves of various compositions of a system during solidification.
- This curve, whatever curve we get here when we study the temperature versus time.
- We can determine the freezing point of the component in the mixture and also the euvectic point.
- How to construct the phase diagram using cooling curves.
- We have studied the simple eutectic system, so here we can see that the phase diagram is live like this.
- We will take up different compositions of A and B and try to plot the cooling curve for each of these compositions.
- We the initially if we when we when it is in the liquid form and we start cooling it, there is just cooling will happen and we don't see anything specific, but when it reaches the melting point or the freezing point, then solid A starts to form.
- And we know that at the freezing point the temperature gets arrested till.

Example or application from the lecture

- Now a very common example for a simple eutectic system would be the lead silver system.
- Let us understand these curves ME as well as NE.

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EV0000214 · EV0000215 · EV0000216 · +1 source units in pack.json

Core explanation · 20:00–24:59

all the liquid has got converted into solid and that is what we see because this is the this is the time axis.

Key details

- And then again when all this liquid has got converted into solid, we find that again cooling is going to happen.
- Say this is the composition and what are we doing here for cooling curve?
- We take this composition heat it to a high temperature beyond the.
- A high temperature when we see that the liquid now the mixture is in the liquid form, it's in the melt state.
- Then we start cooling it.
- When we cool it initially, we don't see anything, there is just a reduction in temperature.
- But the moment it reaches this point.
- We find that the first crystal of A will start to form.
- And we know whenever there is crystallization, then there is a latent heat of fusion involved.
- Definitely there is going to be a change in the slope and that is where we find a change here and that can be taken as the freezing point of A in the mixture B.
- Then further it keeps on cooling down till we come to the eutectic point.
- So when we come when we are cooling this we reach the eutectic protective point we know that now the temperature again gets arrested because now B is also being precipitated.
- We have three phases in equilibrium, F becomes equal to zero as we learned just now, and therefore the temperature becomes constant.
- This is known as the eutectic temperature.
- Which contains A and B of the same composition as the eutectic mill.
- Then after that, there is just a reduction in the temperature, that is there is just cooling.
- This break will tell us what is the freezing point of A in when B is added to it and this point also tells us the eutectic temperature.
- Similarly, if we take another composition, say this composition here, we find that when we melt it and start cooling it, now the freezing will happen at a lower temperature because we are we are here we have added more of B.
- If you look here, this is we have added.
- Added more of B into this, and therefore the freezing point will be much lower, and that's exactly what we see here.
- The first break now when the.
- A pure A will now start coming out of the system, starts precipitating out, we will get the freezing point of A in the mixture, and now this will happen at a much lower temperature because the freezing point has decreased further due to more of B in the mixture.
- Then after that, because of this freezing of A, we find a change in the slope, and that is what helps us to determine the freezing point.
- And then further, when we go down again, we see an arrest in the temperature, and that is the eutectic temperature.
- Composition or discomposition, the eutectic temperature is always constant.
- Further, there is just cooling.
- What is happening here?
- The A now is changing is crystallizing, it is changing from the liquid to the solid, so there is a.

- Arrest in temperatures till all of the liquid has got converted to solid and then there is cooling.
- And then coming to the next point, we have added more of A into B, and therefore we find there is a decrease in the freezing point that.
- Can be noted from the cooling curve where the first crystal of B starts appearing, there is a change in the slope.
- The slope was like this, now suddenly there's a change in the slope.
- Wherever that inflection point, that change in the slope happens, that will be the freezing point of B.

Example or application from the lecture

- Now let us take up the next composition.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000218 · EV0000219 · EV0000220 · +1 source units in pack.json

Core explanation · 25:00–29:59

when A is added into the mixture and then further cooling comes when it reaches the eutectic point.

Key details

- Again, so whether you approach from this side or you approach from this side the eutectic temperature.
- You are going to get the same value of utectic temperature.
- And similarly, if we take this now the eutectic point itself, that would be very interesting because what would happen now when you take this eutectic mixture and you heat it, we find that.
- Initially nothing happens, there's just cooling and the temperature just keeps falling down till we reach this point.
- The because we are at the eutectic point A and B both will start precipitating out and therefore we have three phases and this behaves so if you behave like a pure compound and we get the temperature arrest over here till the eutectic melt has completely converted into the two solids and the composition remains the same.
- The eutectic melt and the solids which are formed have the same composition.
- Therefore we find this arrest in temperature and then there is for the cooling.
- But here we find that there is you can know the utectic temperature.
- And we already know what is the eutectic temperature because that remains constant throughout.
- we can find out wherever these two lines curves meet that would be the eutectic composition.
- The lead silver system shows a simple eutectic diagram and in metallurgy there is a process known as the Pattenson's process.
- Which is used for desilverization of Argentiferous lead.
- The process of heating Argentiferous lead containing a very small quantity of silver, about 0.1%, and cooling it to get pure lead and a liquid melt which is richer in silver.

- When we start cooling it, when we reach this point here, we find that pure lead will start precipitating out and as we cool it further more and more of lead will precipate out and the liquid is going to get richer in silver.
- We go further cool it and then we see that more and more of the lead will precipitate out and the liquid melt composition now is moving towards the area where there is more of silver.
- The liquid melt is getting richer in silver.
- The lead now that is separating out floats and it is continuously removed by layers.
- The liquid melt now contains 2.6% of silver.
- to a composition where the liqu of the liquid weld which has which is much richer in silver and it has now the composition is two point six percent of.

Example or application from the lecture

- Let us now talk about an application of the phase diagram of a two component system, the lead silver system.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000222 · EV0000223 · EV0000224 · +1 source units in pack.json

Core explanation · 30:00–34:59

Natural ore which contains such a small quantity of silver can be made richer in the percentage of silver.

Key details

- So, in the next class we are going to talk about a more complicated system, that is the iron carbon system.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000226 · EV0000227 · EV0000228 · +1 source units in pack.json

13 Computational chemistry: Scope, computational methods, electronic structure determination, AI-enhanced computational chemistry - Part I

Study focus: Computational chemistry: Scope, computational methods, electronic structure determination, AI-enhanced computational chemistry - Part I. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment.

EV0000230 · EV0000231

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session, let us learn Computational Chemistry Introduction, Computational Methods and Applications.

Key details

- Computational chemistry is a multidisciplinary area.
- It involves maths, physics, and also chemistry basic information.
- Computational chemistry is a branch of chemistry that uses mathematical algorithms, statistics, and computer simulations to study and solve chemical problems.
- It does not directly replace experimental chemistry, but it complements predicting molecular structure, properties and reactions, explaining experimental observation.
- reducing the cost and time of the experiment.
- Computational chemistry refers to a collection of techniques used to investigate chemical problems with the aid of computers.
- Rather than relying on laboratory experiments with chemicals, it applies the principles of theoretical chemistry through efficient computer program to predict and calculate molecular properties.
- The earliest theoretical study was done in this field.
- The experiment was carried out by Walter Hitler and Fritz Linden in 1927.
- In the year nineteen ninety eight, Nobel Prize is awarded in chemistry to Walter Conn and John A Popel.
- For the development of density functional theory, as well as for the contributions to the development of computational methods in quantum chemistry, which greatly advanced under practical applications of theoretical models.
- In the year 2013, the Nobel Prize was awarded in chemistry for the development of computer based methods and simulate.
- Complex chemical system enabling a deeper understanding of molecular process.
- Objectives of computational chemistry.
- It predicts and estimates the structure, stability, reactivity, and properties of molecules.
- To provide reliable, accurate, and compressive information at anatomic level.
- The advantages of computational chemistry are calculations are easy to perform than carrying out the experiments in the laboratory.
- Calculations are less costly whereas experiments are becoming more expensive because not using any chemical molecule in the calculation process, so it is less costly.
- Calculations are safer than performing the experiment in the lab.
- These are the advantages of computational chemistry.
- Scope of computational modeling.
- The computational modeling can be used to accurately predict physical and chemical properties of molecule such as bond length, bond angles, humor and lumo interactions.
- It can help in predicting the spectr of ultraviolet visible and IR spectroscopy, as well as proton nuclear magnetic resonance spectroscopy.
- And also able to understand thermodynamic and kinetic properties.
- According to thermodynamic properties are stable and unstable.

- Whether the system is stable or unstable with respect to the chemistry molecule, whether the molecule is stable or highly unstable.
- This can be determined thermodynamically.
- Lability means immediately it can undergo reaction.
- Inertness means certain extent it is stable, higher the activation energy of the molecule.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000230 · EV0000231 · EV0000232 · +1 source units in pack.json

Core explanation · 05:00–09:59

The electronic structure and all unmechanical properties can be determined by using computational modelling.

Key details

- What we need to understand in the reaction, one can easily understand the activation energy.
- After understanding the activation energy,.
- At what temperature a reaction can be carried out?
- What is the pressure?
- Whether it requires to use the catalyst or not.
- All these can be easily determined by using computational tool.
- To know the underlying reaction mechanism and the kinetics of a reaction mechanism, computational chemistry plays a vital role in understanding.
- What is the temperature, pressure, or catalyst is required or not to carry out the experiment in the laboratory.
- And also it helps in material design.
- In the by using computational modeling enables the rational design of new material with desired properties such as electrical conductivity, thermal stability, optical properties, magnetic properties.
- and other properties can be designed easily with the help of computational tool.
- This leads to the development of advanced material for energy production, storage application, especially in the batteries, capacitor, supercapacitor, electronics, and much more applications.
- This is very, very important one in the pharmaceutical industry in the determination of drug.
- Computational chemistry plays a pivotal role in design and the screening of potential drug candidates by simulating the interaction between the drug molecule and biological target, particularly in case of.
- The cancer cells in a particular part of the DNA or an RNA molecule affected by cancer cells, where by using this computational model, one can easily discover a target molecule for the cancer cell.
- develop in the human system.
- That is why in the drug discovery computational model is very much useful, especially structure activity relationship of the drug molecule can be easily determined.

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14 Computational chemistry: Scope, computational methods, electronic structure determination, AI-enhanced computational chemistry - Part III

Study focus: Computational chemistry: Scope, computational methods, electronic structure determination, AI-enhanced computational chemistry - Part III. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment.

EV0000238 · EV0000239

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session we learn electronic structural determination by taking an example of ethylene molecule.

Key details

- Ethylene molecule is the simplest algae having carbon double bound carbon.
- In the determination of electronic structure, the process of finding the arrangement of electrons and atoms and molecules, it is important because chemical bonding reactivity and properties depending on electronic configuration.
- It can be achieved by using theoretical models, computational methods, and spectroscopic techniques.
- You want understood about the iodization and bonding.
- An ethylene molecule is simplest molecule.
- Each carbon atom is sp2 iperized.
- Three sp2 orbitals, two Cs sigma bonds, and one carbon single bond carbon.
- One unibrodized p orbital per carbon.
- Per carbon forms pi bond.
- One sigma bond and one pi bond.
- Description Sigma system means where there is mutual sharing of electrons between two atoms leads to the formation of sigma bond.
- Hence it is called a strong localized framework bonds, call it as sigma system.
- Pi bond is formed where there is sideways overlapping between p orbitals leads to the formation of pi bonding.
- It can be represented as pi bonding and pi star antebonding orbitals.
- Totally it is called as pi system.
- This is the basis for chemical reactivity.

- In the computational determination, some steps are involved.
- The first step is to define the system with respect to the molecule, atom, and geometry.
- Second one, select basis set.
- It is nothing but mathematical functions.
- We have to apply method either at in Gio.
- or density functional theory or semi empirical method.
- Solve the equity get wave function or electron density.
- Visualize electronic structure, orbital diagram, density maps, frontier orbitals like Homo and Lomo.
- Applications in spectroscopy.
- Photoelectron spectroscopy measures sigma and pi energies.
- Usually in ultraviolet visible spectroscopy, sigma to sigma star transition, pi to pi star, pi to n.
- And SIMO N transition takes place in particular.
- Pi to pi star transition takes place between six one sixty five to one eighty nanometer.
- Characteristic stretching frequency is observed for carbon double bond carbon at 1650 reciprocome centimeter.
- In the same way, proton NMR spectroscopy, we will measure proton change.
- In general application, of computational tool.
- It helps in predicting the molecular structure and it helps in understanding electrophilic addition reactions, molecular geometry, bonding nature,.
- And spectroscopic transitions.
- It provides Homo Lomo insights for the reactivity and study reaction mechanism.
- Here one cell explanatory video is playing.
- All of you observe carefully.
- We have to use Goggian software is a powerful tool to understand reactivity and humor interactions.
- All of you observe carefully all these things in the self-explanatory video.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000238 · EV0000239 · EV0000240 · +1 source units in pack.json

Core explanation · 05:00–09:59

[ASR low confidence; check audio].

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

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15 Computational chemistry: Scope, computational methods, electronic structure determination, AI-enhanced computational chemistry - Part II

Study focus: Computational chemistry: Scope, computational methods, electronic structure determination, AI-enhanced computational chemistry - Part II. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment.

EV0000246 · EV0000247

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session, we learn methods of computational chemistry.

Key details

- There are two main methods depending on starting point theory.
- One is called as a classical method, and another one is called as quantum chemistry method.
- In this method, we are using Newton mechanics to model molecular system.
- At the same time,.
- The sum of all the energies of attractive and repulsive forces between atoms in the structure.
- Namely molecular dynamics and molecular mechanics.
- In case of quantum chemistry methods,.
- This uses quantum mechanics to model the molecular system.
- It is based on the arrangement of electrons in molecule and the interaction of these electrons with electron and the nuclei of another molecule.
- And three important methods are there in the quantum chemistry based on this semi empirical methods, Abin show method and density functional theory.
- First, particularly the molecular mechanics as well as molecular dynamics, followed by this, and we have to learn another three methods, namely semi-empirical methods eventually methods, density functional theory in detail.
- What is molecular mechanics?
- Molecular mechanics is a computational method used to study and predict the structure, stability, and properties of molecules using classical physics rather than quantum mechanics.
- It is also based on the ball and spring model.
- It's too fast like a very large molecules like steroids can be optimized in seconds and almost geometry optimization by using molecular mechanics in the classical methods.
- The another technique is called molecular dynamics.
- What is this molecular dynamics?
- Molecular dynamics is a computer simulation technique in which time dependent behavior of a molecular system is studied by solving Newton's equation of motion of atoms and molecules.

- Molecular dynamics is a molecular mechanism program and designed to mimic the movement of atoms within a molecule, again based on the law of motion of molecule, based on the molecular structure.
- Molecular dynamics can be carried out on a molecule to generate different conformations which are energy minimizing minimization.
- I hope you people have understood in the lower standards.
- Take butane molecule.
- Butane molecule is having different conformations.
- What are these conformation?
- The spati arrangement of methyl group in the plane.
- In the semi-empirical form methods, particularly in the quantum mechanics, we do use semi-empirical method.
- What is the semi-empirical method?
- What is the definition?
- Semi-empirical methods are quantum chemistry approaches that simplify the Schrödinger wave equation by making approximation and incorporating experimental data, particularly empirical parameters are used.
- Usually semi-empirical methods values lie between av and show methods, completely quantum, very accurate, but expensive.

Example or application from the lecture

- Let us understand what is classical method.
- We need to consider atom as a spare or ball, and bonds is nothing but a spring, and electrons are ignored here.
- Total potential energy of a molecule is given by.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000246 · EV0000247 · EV0000248 · +1 source units in pack.json

Core explanation · 05:00–09:59

In semi empirical calculations are based on completely Schrödinger equations.

Key details

- It represents a middle road between the mostly qualitative results available from molecular mechanics and the high computationally demanding quantitative results from adventure methods, capable of calculating transition state and excited states.
- What is abinshoe method?
- First and foremost thing abinshoe is the Latin word.
- It means from the beginning.
- What is the definition of events show calculation or events method?
- In computational chemistry, abinshow methods are first principles, quantum, mechanical calculation, and do not use empirical parameters.
- Everything is derived from solving the Schrödinger equation using fundamental constants like electron charge, Planck's constant, etc.

- In the F and Shoe method as already discussed based on the Schrödinger equation, it describes among other things how the electrons in a molecule behave.
- The wave function is a mathematical function that can be used to calculate electron distribution.
- Most accurate quantum chemical techniques, but computationally demanding, they serve as gold standard for validating other computational methods.
- What is the definition for this?
- Density functional theory is a quantum mechanical method used to investigate the electronic structure, especially ground state properties of atoms.
- molecules and solids instead of solving for the mini electron wave function because it becomes very complex.
- DFT works with electron density as the central variable.
- Unlike the other two methods, DFT does not calculate conventional wave function, but rather derives the electron distribution directly.
- Density functional calculations are usually faster than adio.
- but slower than empirical methods.
- In a simpler manner, DFD is a quantum method using electron density instead of wave functions.
- Balances, accuracy, and computational efficiency.
- Computational chemistry is very much essential nowadays.
- We are going to use in the determination of electronic structure.
- And predictions, energy minimization.
- And we can find out the transition structure and reaction parts, molecular docking, protein coupling, protein interaction, protein ligand interactions.
- All these are useful in developing particular molecule especially in the cancer drug anti-cancer drug molecule.
- stagger, anti-gauche, gauche, and all these confirmation what is the stabilization energy that can be calculated by using computational chemistry.
- Thermochemistry Usually in thermochemistry two reaction takes place one is exothermic and endothermic reactions and the heat of reactions can be easily calculated, energy of activation of the molecule can be calculated using computational chemistry.
- Calculation of many other molecular physical and chemical properties can be explored by using computational chemistry methods.
- Computational chemistry plays a vital role in progenitic world.

Example or application from the lecture

- So let us understand ab in show method.
- Let us understand what is density functional theory.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000250 · EV0000251 · EV0000252 · +1 source units in pack.json

Core explanation · 10:00–14:59

particularly in advancing our understanding of chemical system and accelerating scientific discovery.

Key details

- It has one of the important advantages is that cost effectiveness, efficiency, safety, accessibility, versatility, and accuracy make it indispensable tool in modern research and development.
- Computational chemistry is not a replacement from the lab.
- Because we need to.
- Perform the experiment to get a desired molecule.
- But this computational chemistry will help us whether we are going to get desired molecule or not before conducting experiment in the lab.
- That is what computational chemistry is not a replacement for experimental studies, but plays an important role in enabling chemists to design and develop the design chemicals and the product molecule.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000254 · EV0000255 · EV0000256 · +1 source units in pack.json

16 Source slide deck: Born Oppenheimer approximation

Study focus: Source slide deck: Born Oppenheimer approximation. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000312 · EV0000313

SHORT-NOTE RECAP

Study notes · Source slide deck: Born Oppenheimer approximation

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · ENGINEERING CHEMISTRY (UE26CY151 A,B)

- Department of Science and Humanities

Slide 2 · ENGINEERING CHEMISTRY

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 3 · ENGINEERING CHEMISTRY

- Translational energy (Etrans): by virtue of translatory motion of the molecule
- Rotational energy (Erot) : by virtue of rotation of a molecule about its centre of gravity
- Vibrational energy (Evib): by virtue of periodic diaplacement of the atoms of a molecule about its equilibrium position

Slide 4 · ENGINEERING CHEMISTRY

- Total energy of a molecule is the sum of translational, rotational, vibrational and electronic energies, i.e.
- Translational energy is negligibly small. Hence Born-Oppenheimer approximation can be written as
- Because the electronic energy levels are much farther apart than the vibrational energy levels which are again much farther apart than the rotational energy levels

Slide 5 · ENGINEERING CHEMISTRY

- Energy level diagram for a diatomic molecule
- Source:http://hyperphysics.phy-astr.gsu.edu/hbase/molecule/molec.html

Slide 6 · Department of Science and Humanities

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

EV0000312 · EV0000313 · EV0000314 · +15 source units in pack.json

17 Source slide deck: Experiential learning Based on Phase Diagram

Study focus: Source slide deck: Experiential learning Based on Phase Diagram. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000663 · EV0000664

SHORT-NOTE RECAP

Study notes · Source slide deck: Experiential learning Based on Phase Diagram

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · ENGINEERING CHEMISTRY (UE26CY151 A,B)

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Slide 2 · ENGINEERING CHEMISTRY

- Experiential Learning – UNIT 1 (supplementary material)
- For the determination of equilibrium conditions between
- The study of the cooling curves of various compositions of a system during solidification

Slide 3 · ENGINEERING CHEMISTRY

- Source:http://www.mchmultimedia.com/PhysicalChemistry-help/clientstories/study-tips/digging-into-phase-diagrams-cooling-curves.html
- For a mixture of solids: When crystallisation of one of the components starts ,cooling curve exhibits a break. The temperature decreases continuously until the eutectic point is reached. Now the temperature remains constant, till the completion of solidification.
- For pure solid: Initially the rate of cooling is continuous from ‘a’ till the point ‘b’ where solid is begins to appear. Then the temperature remains constant until the liquid is fully solidified and solidification completes at the point ‘c’.

Slide 4 · [Image text, OCR]

- Department of Science and Humanities

EV0000663 · EV0000664 · EV0000665 · +9 source units in pack.json

18 Historical question-source pages · unit mapping candidates

Study focus: Historical question-source pages · unit mapping candidates. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000675 · EV0000676

SHORT-NOTE RECAP

Study notes · Historical question-source pages · unit mapping candidates

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Archive page 12 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.8333; source text/OCR may be partial. ENGINEERING CHEMISTRY Module I- COMPUTATIONAL CHEMISTRY Computational methods: There are two main methods depending on the starting point theory: Classical method: Quantum chemistry method : • These use Newton mechanics to model molecular system. • Atom – Sphere, Bonds – Springs and Electrons are ignored. • Total potential energy of a molecule is given by sum of all the energies of attractive and repulsive forces between atom in structure. ☒ These use quantum mechanics to model the molecular system. ☒ It is based on arrangement of electrons in molecules and interaction of these electrons with electron and nuclei of another molecule. ☒ We consider that electron and all other particle exhibits both particle and wave nature. i. Molecular mechanics ii. Molecular dynamics i. Semi empirical methods. ii. Ab initio methods iii. Density functional theory Ref. Book: Introduction to computational chemistry by Frank Jensen, Wiley publications Classical Method Quantum Chem Method

Archive page 2 from 10__1vFS6E1bKV.pdf; unit match candidate score 1.0; source text/OCR may be partial. (4.0 Marks) (2.0 Marks) (5.0 Marks) 1.d. For HBr molecule: (4M) (i) The rotational spectrum shows the first line at 17.19 cm . Calculate the bond length of the HBr molecule (use rigid rotor model). (ii) The vibrational spectrum shows fundamental and the first overtone at 2559.08 cm and 5027.54 cm respectively. Calculate the anharmonicity constant for the molecule (use the anharmonic oscillator model). (1 amu = 1.66 x 10 kg, h = 6.627 x 10 J s, c = 3 x 10 cm s , N = 6.023 x 10 mol , gram molar mass of H = 1.0 and Br = 79.9). 1.e. Give the two step synthesis of alpha Aluminium oxide in solution combustion method (2M) 2.a. i.In the given phase diagram of water; in fusion curve OC, identify the number of phase, variance and reason out the negative slop with variation of melting point of ice with pressure. (3M) ii. In the following phase equilibrium, identify the number of phases and components (2M) a. Ammonium chloride undergoes thermal decomposition NH4Cl(s) ⇌ NH3 (g) + HCl (g) (closed system) b. CaCO (s) ↔CaO(s) + CO (g) (Closed System) –1 –1 –1 –27 –34 10 –1 A 23 –1 3 2

Archive page 2 from 11__1zfeA5C-J7.pdf; unit match candidate score 0.6667; source text/OCR may be partial. (2.0 Marks) (6.0 Marks) (6.0 Marks) (6.0 Marks) 1.d. What is the reducing agent used during the' synthesis of α - Al O nano metal oxide by solution combustion method' experiment? Why should the reaction be carried out in open containers with minimum quantity of redox mixtures in the lab? 2.a. (i) What is the maximum % of carbon taken along the composition axis in the phase diagram of two component Fe-C system? Why? (2marks) (ii) Why is reduced/condensed phase rule used for two component Pb-Ag and Fe-C systems? (iii) Which are the three phases in equilibrium at the eutectic point in the two component Pb - Ag system? What is the degree of freedom of the system at this point? 2.b. (i) For the cell represented below, calculate the E . Pt/H (10 atm)/H (0.3M) / H (3atm)/Pt (2 marks) (ii) A glass electrode is coupled with saturated calomel electrode to measure unknown pH. The cell potentials measured are 0.215V and 0.385V in contact with a solution of pH = 7 and with solution of unknown pH respectively. Calculate the pH of unknown solution. Given E = 0.244V 2.c. (i) For the cell represented below- Ag/AgCl/Cl (0.1M) // Fe (0.02M), Fe (0.1)/ Pt (i) Write the overall cell reaction. (2 marks) (ii)Calculate E and E at 298 K (4 marks) (Given: The standard reduction potential of E Fe /Fe = 0.77 V and that of E = 0.222 V, R = 8.314 J/K/mol, F =96500 C/mol) 2 3 cell 2 + 2 SCE -1 2+ 3+ oCell Cell 0 3+ 2+ 0Ag/AgCl

Archive page 1 from 1__1rBrYYqw3s.pdf; unit match candidate score 1.0; source text/OCR may be partial. (6.0 Marks) (6.0 Marks) (4.0 Marks) PES University, Bengaluru (Established under Karnataka Act 16 of 2013) END SEMESTER ASSESSMENT (ESA) - Jan - June 2024 UE23CY151B - Engineering Chemistry Total Marks : 100.0 1.a. For a rigid rotor diatomic molecule: (i) Draw the energy level diagram up to J=5. (ii) Draw the rotational spectrum. (iii) Calculate the moment of inertia and the distance between two spectral line in the spectra if B = 10.56 cm . (Given : c = 3 x 10 m/s, π =3.14, h = 6.625x10 Js respectively) 1.b. (i) The force constant of HF molecule showing simple harmonic oscillation at 970 N/m. Calculate the oscillation frequency (cm) and zero point energy. (ii) Why does the spectrum shows only one line in simple harmonic oscillator? (Given: c =3 x 10 m/sec, π=3.14, N = 6.02 x 10 , 1 amu =1.66x10 kg, molar mass of H =1.008 amu, F = 18.99 amu). 1.c. Derive phase rule for a system of C componens and P phases which are in equilibrium with each other. -1 8 -34 -1 8 23 -27

Archive page 1 from 2__11pj9jsL0V.pdf; unit match candidate score 1.25; source text/OCR may be partial. (6.0 Marks) (6.0 Marks) (4.0 Marks) PES University, Bengaluru (Established under Karnataka Act 16 of 2013) END SEMESTER ASSESSMENT (ESA) - Jan - June 2024 UE23CY151B - Engineering Chemistry Total Marks : 100.0 1.a. For a rigid rotor hetero diatomic molecule (AX), The first line in the pure rotational spectrum corresponds to 21.18 cm . Derive an expression for the moment of inertia and Calculate the moment of inertia and bond length of the AX molecule. (Given: Atomic masses of A = 1.008 amu and X = 35.45 amu, Avagadro's number = 6.023 x 10 , Speed of light = 3 x 10 cm/s, π = 3.14, Planck's constant = 6.627 x 10 Js, 1 amu = 1.66 x 10 kg). 1.b. i). The radiation absorbed by CO during a vibrational transition is 2168 cm . Calculate ground state vibrational energy and force constant. (Simple harmonic oscillator model Given c = 3×10 cm/s, Planck's constant = 6.625×10 Js, N = 6.02×10 ,1 amu = 1.66 ×10 kg, atomic masses: C=12, O=15.999). ii). Write a note on the failure of a simple harmonic oscillator model. 1.c. Justify the Franck-Condon principle by drawing the vibrational-electronic transitions in the following cases and draw the spectra arising from them: 1. When the internuclear distances in the ground and excited states are the same. 2. When the internuclear distance in the excited state is much larger than that in the ground state. -1 23 10 - 34 -27 -1 10 -34 23 -27

Archive page 1 from 3__1MwY1CTl_j.pdf; unit match candidate score 1.0; source text/OCR may be partial. 3/25/25, 3:51 PM ESA - Dec 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 1/7 (5.0 Marks) (5.0 Marks) (4.0 Marks) PES University, Bengaluru (Established under Karnataka Act 16 of 2013) END SEMESTER ASSESSMENT (ESA) - Dec 2024 UE23CY151B - Engineering Chemistry Total Marks : 100.0 1.a. Derive the expression for the moment of inertia for a diatomic rigid rotor 1.b. Briefly describe the Born-Oppenheimer aproximation and draw the ground and excited electronic states of a diatomic molecule showing the rotational and vibrational energy levels (using anharmonic oscillator model). 1.c. The force constant of CO is 1840 Nm . Calculate the oscillation frequency and wave number in cm . (1 amu = 1.66 x 10 kg, c = 3 x 10 ms , Gram molar mass of C = 12.000 and O = 15.9994) -1 -1 -27 8 -1

Archive page 2 from 3__1MwY1CTl_j.pdf; unit match candidate score 0.5833; source text/OCR may be partial. 3/25/25, 3:51 PM ESA - Dec 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 2/7 (8.0 Marks) (3.0 Marks) (5.0 Marks) 1.d. (i) Draw the phase diagram of water, label the areas and the phase boundaries, and calculate degrees of freedom in the areas, phase

boundaries, and the triple point. (4M) (ii) With the help of Clausius-Clapeyron equation, discuss why the melting point curve of the water phase diagram has a negative slope (2M) (iii) What is reduced phase rule and under which situation it is used? (2M) 1.e. State the Beer-Lambert law. Why is ammonia added during colorimetric estimation of copper? (2M + 1M) 2.a. Derive the Nernst equation for a single electrode undergoing the following electrode reaction: M + ne → M n+

Archive page 1 from 4__1eKMFaJ2lx.pdf; unit match candidate score 1.0; source text/OCR may be partial. 3/27/25, 11:47 AM ESA June 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 1/7 (6.0 Marks) (6.0 Marks) (4.0 Marks) PES University, Bengaluru (Established under Karnataka Act 16 of 2013) END SEMESTER ASSESSMENT (ESA) - Jan - June 2024 UE23CY151B - Engineering Chemistry Total Marks : 100.0 1.a. For a rigid rotor diatomic molecule: (i) Draw the energy level diagram up to J=5. (ii) Draw the rotational spectrum. (iii) Calculate the moment of inertia and the distance between two spectral line in the spectra if B = 10.56 cm . (Given : c = 3 x 10 m/s, π =3.14, h = 6.625x10 Js respectively) 1.b. (i) The force constant of HF molecule showing simple harmonic oscillation at 970 N/m. Calculate the oscillation frequency (cm) and zero point energy. (ii) Why does the spectrum shows only one line in simple harmonic oscillator? (Given: c =3 x 10 m/sec, π=3.14, N = 6.02 x 10 , 1 amu =1.66x10 kg, molar mass of H =1.008 amu, F = 18.99 amu). 1.c. Derive phase rule for a system of C componens and P phases which are in equilibrium with each other. -1 8 -34 -1 8 23 -27

Archive page 1 from 6__1H6QH_PVNf.pdf; unit match candidate score 1.0; source text/OCR may be partial. 3/27/25, 11:47 AM ESA June 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 1/7 (6.0 Marks) (6.0 Marks) (4.0 Marks) PES University, Bengaluru (Established under Karnataka Act 16 of 2013) END SEMESTER ASSESSMENT (ESA) - Jan - June 2024 UE23CY151B - Engineering Chemistry Total Marks : 100.0 1.a. For a rigid rotor diatomic molecule: (i) Draw the energy level diagram up to J=5. (ii) Draw the rotational spectrum. (iii) Calculate the moment of inertia and the distance between two spectral line in the spectra if B = 10.56 cm . (Given : c = 3 x 10 m/s, π =3.14, h = 6.625x10 Js respectively) 1.b. (i) The force constant of HF molecule showing simple harmonic oscillation at 970 N/m. Calculate the oscillation frequency (cm) and zero point energy. (ii) Why does the spectrum shows only one line in simple harmonic oscillator? (Given: c =3 x 10 m/sec, π=3.14, N = 6.02 x 10 , 1 amu =1.66x10 kg, molar mass of H =1.008 amu, F = 18.99 amu). 1.c. Derive phase rule for a system of C componens and P phases which are in equilibrium with each other. -1 8 -34 -1 8 23 -27

Archive page 1 from 7__16pEjcSTVO.pdf; unit match candidate score 1.0; source text/OCR may be partial. (6.0 Marks) (6.0 Marks) (6.0 Marks) PES University, Bengaluru (Established under Karnataka Act 16 of 2013) END SEMESTER ASSESSMENT (ESA) - Jan - May 2024 UE22CY151B - Engineering Chemistry Total Marks : 100.0 1.a. (i) State Born Openhemier approximation? (ii) What does an appearance of continnum in an electronic spectra lead us to conclude. 1.b. (i) Derive an expression for the moment of inertia for a diatomic molecule (rigid rotar model). (ii) Why is carbon dioxide infrared active but microwave inactive? 1.c. (I) For a HBr molecule undergoing simple harmonic motion the vibrational spectrum shows a fundamental frequency at 2649 cm . determine force constant of HBr Given the reduced mass of HBr is 1.639x10−27 Kg and c= 3x10 m/s. (ii) What are three dimensional nanomaterials? Give one example. -1 8

Archive page 2 from 7__16pEjcSTVO.pdf; unit match candidate score 0.5; source text/OCR may be partial. (2.0 Marks) (6.0 Marks) (6.0 Marks) (6.0 Marks) 1.d. Why should the synthesis of α Al O be done in open container with a minimum quantity of reactants and in a well ventilated set up? 2.a. (i) Give Gibbs Phase rule and explain the tems involved. (ii) At triple point of one component water system, what are the phases in equilibrium and the the degree of freedom. 2.b. (i) For the cell represented below, calculate the E Au/Au (0.05M) // Au (0.12 M)/ Au (ii) When a glass electrode is combined with saturated calomel electrode and immersed in a solution of pH 7 the E recorded was 0.215V, when the same electrode assembly was immersed in a solution of unknown pH the E was recorded as 0.385V. Using the data determine the pH of unknown solution. Given E is 0.24V. 2.c. For the below mentioned cell representation- Fe/Fe (0.05M) //Ag (0.1M)/Ag Given : E = −0.44V ; E = 0.8 V (i) write the anodic, cathodic and net cell reaction. (ii) Calculate the E and E 2 3 cell 3+ 3+ cell cell SCE 2+ + 0Fe / Fe +2 0Ag /Ag + 0cell cell

Archive page 2 from 9__14fBQb1-hv.pdf; unit match candidate score 0.5; source text/OCR may be partial. 11/8/24, 1:02 PM ESA - DEC 2023 - UE22CY151B/UE22CY151A (set- 1) about:blank 2/9 (4.0 Marks) (2.0 Marks) (5.0 Marks) 1.d. The first line in the pure rotational spectrum of HCl appears at 21.18 cm-1 . Calculate the bond length of the molecule. Given atomic masses of H and Cl are 1.008 amu and 35.45 amu respectively. (1 amu = 1.66 x 10-27 kg, c = 3 x 108 ms-1, h = 6.6x 10-34 Js) 1.e. Mention the precautions to be taken while synthesis of alfa-Al2O3 nano metal oxide via compustion method 2.a. The study of phase diagram of a 2-component system has formed the basis for Pattinson's process for the desilverisation of argentiferous lead, illustrate with the phase diagram of lead-silver systerm.

Archive page 1 from Chem-RR[2025]__1xN30tqzNj.pdf; unit match candidate score 1.3333; source text/OCR may be partial. 3/25/25, 10:57 AM ESA - Jan 2025 - UE24CY151A (set- 1) about:blank 1/7 (6.0 Marks) (7.0 Marks) (6.0 Marks) PES University, Bengaluru (Established under Karnataka Act 16 of 2013) END SEMESTER ASSESSMENT (ESA) - Jan-Feb UE24CY151A - Engineering Chemistry Total Marks : 100.0 1.a. i) Why HI molecule is IR active but H and I are IR inactive? (1M) ii) The first spectral line in the rotational spectrum of HI molecule appears at 13.022 cm . Calculate the bond length. (3M) (Given: c = 3 x 10 m/s, 1amu=1.66 x 10 kg, gram molar mass of H =1.0 amu and I=126.9 amu, π=3.14, Planck's constant = 6.627x10 Js) iii) Draw an energy level diagram of ground and excited electronic energy states of a diatomic molecule and show vibrational and rotational energy levels within them. (2M) 1.b. i) Using harmonic oscillator model calculate zero point vibrational energy of NO molecule given that the force constant of NO is 1530 N/m. (4M) ii) Draw the vibrational energy level diagram for the above model and draw the corresponding spectra obtained from the transition. (2M) iii) Calculate the energy of v=3 and v=4 level. (1M) (Given: c = 3x10 m/s, 1amu=1.66x10 kg, gram molar mass of N =14.0 amu and O=16 amu, π=3.14, Planck's constant = 6.627x10 Js) 1.c. i) Draw a neat labeled phase diagram of water system (mention the areas, lines, triple point, critical point, pressure, temperature) and calculate the degree of freedom along the freezing point curve. (4M) ii) What is condensed phase rule and why is it applied to the two component lead-silver system? (2M) 2 2 -1 8 -27 -34 8 -27 -34

EV0000675 · EV0000676 · EV0000677 · +10 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Official unit catalogue	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/units.json	Official course/unit title and index.
Interaction of radiation with matter transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/001_Interaction_of_radiation_with_matter__1eb05f6c3f230eec.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Interaction of radiation with matter transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/001_Interaction_of_radiation_with_matter__1eb05f6c3f230eec.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Interaction of radiation with matter transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/001_Interaction_of_radiation_with_matter__1eb05f6c3f230eec.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
Interaction of radiation with matter transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/001_Interaction_of_radiation_with_matter__1eb05f6c3f230eec.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Electromagnetic spectrum transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/002_Electromagnetic_spectrum__d84acfbd19ed2af4.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Electromagnetic spectrum transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/002_Electromagnetic_spectrum__d84acfbd19ed2af4.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Electromagnetic spectrum transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/002_Electromagnetic_spectrum__d84acfbd19ed2af4.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9997; only the JSON segmentation supplies the displayed timestamps.
Electromagnetic spectrum transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/002_Electromagnetic_spectrum__d84acfbd19ed2af4.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Rotational spectroscopy transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/004_Rotational_spectroscopy__cda73f2030eb1a62.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Rotational spectroscopy transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/004_Rotational_spectroscopy__cda73f2030eb1a62.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Rotational spectroscopy transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/004_Rotational_spectroscopy__cda73f2030eb1a62.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9997; only the JSON segmentation supplies the displayed timestamps.
Rotational spectroscopy transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/004_Rotational_spectroscopy__cda73f2030eb1a62.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Vibrational spectroscopy(harmonic oscillator) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/005_Vibrational_spectroscopy_harmonic_oscillator__bd1d3b496879b403.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Vibrational spectroscopy(harmonic oscillator) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/005_Vibrational_spectroscopy_harmonic_oscillator__bd1d3b496879b403.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Vibrational spectroscopy(harmonic oscillator) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/005_Vibrational_spectroscopy_harmonic_oscillator__bd1d3b496879b403.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
Vibrational spectroscopy(harmonic oscillator) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/005_Vibrational_spectroscopy_harmonic_oscillator__bd1d3b496879b403.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Vibrational spectroscopy(anharmonic oscillator) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/006_Vibrational_spectroscopy_anharmonic_oscillator__4fe4d43ab2b66d24.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Vibrational spectroscopy(anharmonic oscillator) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/006_Vibrational_spectroscopy_anharmonic_oscillator__4fe4d43ab2b66d24.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Vibrational spectroscopy(anharmonic oscillator) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/006_Vibrational_spectroscopy_anharmonic_oscillator__4fe4d43ab2b66d24.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.
Vibrational spectroscopy(anharmonic oscillator) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/006_Vibrational_spectroscopy_anharmonic_oscillator__4fe4d43ab2b66d24.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Electronic spectroscopy - Franck-Condon principle (Part II) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/007_Electronic_spectroscopy_-_Franck-Condon_principle_Part_II__70e8e6f497126f7d.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Electronic spectroscopy - Franck-Condon principle (Part II) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/007_Electronic_spectroscopy_-_Franck-Condon_principle_Part_II__70e8e6f497126f7d.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Electronic spectroscopy - Franck-Condon principle (Part II) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/007_Electronic_spectroscopy_-_Franck-Condon_principle_Part_II__70e8e6f497126f7d.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9987; only the JSON segmentation supplies the displayed timestamps.
Electronic spectroscopy - Franck-Condon principle (Part II) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/007_Electronic_spectroscopy_-_Franck-Condon_principle_Part_II__70e8e6f497126f7d.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Electronic spectroscopy(Progression) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/007_Electronic_spectroscopy_Progression__6dd27636f794caa1.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Electronic spectroscopy(Progression) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/007_Electronic_spectroscopy_Progression__6dd27636f794caa1.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Electronic spectroscopy(Progression) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/007_Electronic_spectroscopy_Progression__6dd27636f794caa1.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9996; only the JSON segmentation supplies the displayed timestamps.
Electronic spectroscopy(Progression) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/007_Electronic_spectroscopy_Progression__6dd27636f794caa1.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Phase, Component, Degrees of freedom, Statement of Gibb's phase rule, chemical potential transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__a7da1e88f96b163b.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Phase, Component, Degrees of freedom, Statement of Gibb's phase rule, chemical potential transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__a7da1e88f96b163b.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Phase, Component, Degrees of freedom, Statement of Gibb's phase rule, chemical potential transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__a7da1e88f96b163b.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9982; only the JSON segmentation supplies the displayed timestamps.
Phase, Component, Degrees of freedom, Statement of Gibb's phase rule, chemical potential transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__a7da1e88f96b163b.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Derivation of Gibb's phase rule, Phase diagram of one-component system - water - Part II transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water_-_Part_II__2a5eb65b0585fe28.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Derivation of Gibb's phase rule, Phase diagram of one-component system - water - Part II transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water_-_Part_II__2a5eb65b0585fe28.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Derivation of Gibb's phase rule, Phase diagram of one-component system - water - Part II transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water_-_Part_II__2a5eb65b0585fe28.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9967; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Derivation of Gibb's phase rule, Phase diagram of one-component system - water - Part II transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water_-_Part_II__2a5eb65b0585fe28.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Derivation of Gibb's phase rule, Phase diagram of one-component system - water - Part I transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water_-_Part_I__f611c716baed093a.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Derivation of Gibb's phase rule, Phase diagram of one-component system - water - Part I transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water_-_Part_I__f611c716baed093a.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Derivation of Gibb's phase rule, Phase diagram of one-component system - water - Part I transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water_-_Part_I__f611c716baed093a.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9976; only the JSON segmentation supplies the displayed timestamps.
Derivation of Gibb's phase rule, Phase diagram of one-component system - water - Part I transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water_-_Part_I__f611c716baed093a.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Reduced phase rule, phase diagram of two-component system - Pb-Ag system transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/010_Reduced_phase_rule_phase_diagram_of_two-component_system_-_Pb-Ag_system__8e25eaa5432159a1.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Reduced phase rule, phase diagram of two-component system - Pb-Ag system transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/010_Reduced_phase_rule_phase_diagram_of_two-component_system_-_Pb-Ag_system__8e25eaa5432159a1.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Reduced phase rule, phase diagram of two-component system - Pb-Ag system transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/010_Reduced_phase_rule_phase_diagram_of_two-component_system_-_Pb-Ag_system__8e25eaa5432159a1.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9987; only the JSON segmentation supplies the displayed timestamps.
Reduced phase rule, phase diagram of two-component system - Pb-Ag system transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/010_Reduced_phase_rule_phase_diagram_of_two-component_system_-_Pb-Ag_system__8e25eaa5432159a1.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Computational chemistry: Scope, computational methods, electronic structure determination, AI-enhanced computational chemistry - Part I transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__0671dbe2a6d868d4.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Computational chemistry: Scope, computational methods, electronic structure determination, AI-enhanced computational chemistry - Part I transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__0671dbe2a6d868d4.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Computational chemistry: Scope, computational methods, electronic structure determination, AI-enhanced computational chemistry - Part I transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__0671dbe2a6d868d4.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9907; only the JSON segmentation supplies the displayed timestamps.
Computational chemistry: Scope, computational methods, electronic structure determination, AI-enhanced computational chemistry - Part I transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__0671dbe2a6d868d4.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Computational chemistry: Scope, computational methods, electronic structure determination, AI-enhanced computational chemistry - Part III transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__5a0b9f520a3c63b3.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Computational chemistry: Scope, computational methods, electronic structure determination, AI-enhanced computational chemistry - Part III transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__5a0b9f520a3c63b3.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Computational chemistry: Scope, computational methods, electronic structure determination, AI-enhanced computational chemistry - Part III transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__5a0b9f520a3c63b3.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9860; only the JSON segmentation supplies the displayed timestamps.
Computational chemistry: Scope, computational methods, electronic structure determination, AI-enhanced computational chemistry - Part III transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__5a0b9f520a3c63b3.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Computational chemistry: Scope, computational methods, electronic structure determination, AI-enhanced computational chemistry - Part II transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__74b8d88c6bf2ef3c.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Computational chemistry: Scope, computational methods, electronic structure determination, AI-enhanced computational chemistry - Part II transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__74b8d88c6bf2ef3c.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Computational chemistry: Scope, computational methods, electronic structure determination, AI-enhanced computational chemistry - Part II transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__74b8d88c6bf2ef3c.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9940; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Computational chemistry: Scope, computational methods, electronic structure determination, AI-enhanced computational chemistry - Part II transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__74b8d88c6bf2ef3c.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Slide page text: Interaction of electromagnetic radiation with matter	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/001_Interaction_of_electromagnetic_radiation_with_matter__9d3205fc29225eeb.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Interaction of electromagnetic radiation with matter	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/001_Interaction_of_electromagnetic_radiation_with_matter__9d3205fc29225eeb.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Interaction of electromagnetic radiation with matter (converted PDF; original hash 4c9c45c71266e6a0e44f1edeeaa2f0ff7a2976a7d76a178093c7c5b191877db5)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/001_Interaction_of_electromagnetic_radiation_with_matter__9d3205fc29225eeb.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Electromagnetic spectrum	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/002_Electromagnetic_spectrum__ef06d432886fbdb5.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Electromagnetic spectrum	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/002_Electromagnetic_spectrum__ef06d432886fbdb5.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Electromagnetic spectrum (converted PDF; original hash fae8af7f3abd0f8e1640189a707e1813a92e86f60f6178090157f5495ecbd9ed)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/002_Electromagnetic_spectrum__ef06d432886fbdb5.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Born Oppenheimer approximation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/003_Born_Oppenheimer_approximation__342acd62ba6e1f57.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Born Oppenheimer approximation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/003_Born_Oppenheimer_approximation__342acd62ba6e1f57.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Born Oppenheimer approximation (converted PDF; original hash 01f7b22262b35a5b19fe37907dd488fade2c6b0d34195e39f50256e30e05b682)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/003_Born_Oppenheimer_approximation__342acd62ba6e1f57.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Microwave spectroscopy diatomic rigid rotor model and the rotational spectrum	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/004_Microwave_spectroscopy_-_diatomic_rigid_rotor_model_and_the_rotational_spectrum__416fab5550d2cc51.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Microwave spectroscopy diatomic rigid rotor model and the rotational spectrum	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/004_Microwave_spectroscopy_-_diatomic_rigid_rotor_model_and_the_rotational_spectrum__416fab5550d2cc51.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Microwave spectroscopy diatomic rigid rotor model and the rotational spectrum (converted PDF; original hash fe26fc505240d4a63cbfd4a4728a57f0fe4e56fda0e7888be1d7ed342a585ed1)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/004_Microwave_spectroscopy_-_diatomic_rigid_rotor_model_and_the_rotational_spectrum__416fab5550d2cc51.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: IR spectroscopy diatomic harmonic oscillator model	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/005_IR_spectroscopy_-_diatomic_harmonic_oscillator_model__044502d9a73710b7.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: IR spectroscopy diatomic harmonic oscillator model	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/005_IR_spectroscopy_-_diatomic_harmonic_oscillator_model__044502d9a73710b7.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for IR spectroscopy diatomic harmonic oscillator model (converted PDF; original hash 449aef5af8ed12ee270a3dd50b67af77db157bd304cf802b9d0a7db1012c76f8)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/005_IR_spectroscopy_-_diatomic_harmonic_oscillator_model__044502d9a73710b7.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: IR spectroscopy anharmonic oscillator model	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/006_IR_spectroscopy_-_anharmonic_oscillator_model__ac64a226f45a8e06.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: IR spectroscopy anharmonic oscillator model	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/006_IR_spectroscopy_-_anharmonic_oscillator_model__ac64a226f45a8e06.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for IR spectroscopy anharmonic oscillator model (converted PDF; original hash d8de97dd12513d33d5f25e51c96df34737da95e7c8da27403471afd7c29b952b)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/006_IR_spectroscopy_-_anharmonic_oscillator_model__ac64a226f45a8e06.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Electronic spectroscopy vibrational coarse structure progressions Franck Condon principle	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Electronic spectroscopy vibrational coarse structure progressions Franck Condon principle	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Electronic spectroscopy vibrational coarse structure progressions Franck Condon principle (converted PDF; original hash ab6e8532f77273da103d72afdcbb94b3f20bda0bf1afe42fc4d49febb73bf784)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Phase Component Degrees of freedom Statement of Gibb s phase rule chemical potential	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__eb00184a944a5a23.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Phase Component Degrees of freedom Statement of Gibb s phase rule chemical potential	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__eb00184a944a5a23.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Phase Component Degrees of freedom Statement of Gibb s phase rule chemical potential (converted PDF; original hash e5286 31d79770489e2f8b6b485441c06503be6f69b0dc9fa6db4c18735f96c20)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__eb00184a944a5a23.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Derivation of Gibb s phase rule Phase diagram of one component system water	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Derivation of Gibb s phase rule Phase diagram of one component system water	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Derivation of Gibb s phase rule Phase diagram of one component system water (converted PDF; original hash 72fc1 f0d9f3152dc670b650a3653f9702ffcf97fd1d13ef290d1f8547dd43d5e)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.

Source	Path	Use
Slide page text: Reduced phase rule phase diagram of two component system Pb Ag system	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/010_Reduced_phase_rule_phase_diagram_of_two-component_system_-_Pb-Ag_system__da2fe68080b8e9cb.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Reduced phase rule phase diagram of two component system Pb Ag system	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/010_Reduced_phase_rule_phase_diagram_of_two-component_system_-_Pb-Ag_system__da2fe68080b8e9cb.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Reduced phase rule phase diagram of two component system Pb Ag system (converted PDF; original hash a52eedc2c4ce354e84cf7c3619435ef387196e6a3aa19475ccc063cd5cb89bec)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/010_Reduced_phase_rule_phase_diagram_of_two-component_system_-_Pb-Ag_system__da2fe68080b8e9cb.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Computational chemistry Scope computational methods electronic structure determination AI enhanced computation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Computational chemistry Scope computational methods electronic structure determination AI enhanced computation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Computational chemistry Scope computational methods electronic structure determination AI enhanced computation (converted PDF; original hash bbd5a5fef818c52e5f7de4b010ff65e9086c2fe3947500de1f80f4fbc2e1b37a)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Experiential learning Based on Phase Diagram	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/012_Experiential_learning_Based_on_Phase_Diagram__42cc75d52ba7f0e4.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Experiential learning Based on Phase Diagram	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/012_Experiential_learning_Based_on_Phase_Diagram__42cc75d52ba7f0e4.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Experiential learning Based on Phase Diagram (converted PDF; original hash 4f25bcdae7b386b7085c93346836685287dafdaced1ee62de23f9a750f12b5bc)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/012_Experiential_learning_Based_on_Phase_Diagram__42cc75d52ba7f0e4.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Extracted historical pages: Chem_Extra_Topics__1-HfOeZNwo.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Extra/Chem_Extra_Topics__1-HfOeZNwo.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 10__1vFS6E1bKV.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/10__1vFS6E1bKV.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 11__1zfeA5C-J7.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/11__1zfeA5C-J7.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.

Source	Path	Use
Extracted historical pages: 1__1rBrYYqw3s.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/1__1rBrYYqw3s.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 2__11pj9jsL0V.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/2__11pj9jsL0V.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 3__1MwY1CTl_j.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/3__1MwY1CTl_j.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 4__1eKMFaJ2lx.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/4__1eKMFaJ2lx.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 6__1H6QH_PVNf.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/6__1H6QH_PVNf.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 7__16pEjcSTVO.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/7__16pEjcSTVO.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 9__14fBQb1-hv.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/9__14fBQb1-hv.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: Chem-RR[2025]__1xN30tqzNj.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2025/Chem-RR[2025]__1xN30tqzNj.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.

2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00001	1 covered · 0 excluded · 0 gaps	catalogue: Unit title and index copied from the official units.json catalogue; human content review is pending.
SRC00002	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00003	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...
SRC00004	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00005	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...
SRC00006	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00007	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0059. Sidecar variants compare...
SRC00008	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00009	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0059. Sidecar variants compare...
SRC00010	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00011	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...
SRC00012	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00013	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...
SRC00014	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00015	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...
SRC00016	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00017	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...
SRC00018	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00019	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0061. Sidecar variants compare...
SRC00020	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00021	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0061. Sidecar variants compare...
SRC00022	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00023	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare...
SRC00024	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00025	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare...
SRC00026	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00027	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...
SRC00028	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00029	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...
SRC00030	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00031	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...
SRC00032	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00033	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...
SRC00034	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00035	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...
SRC00036	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00037	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...
SRC00038	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00039	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare...
SRC00040	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00041	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare...
SRC00042	7 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00043	7 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...
SRC00044	7 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00045	7 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...
SRC00046	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0033. Sidecar variants compare...
SRC00047	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0033. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00048	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0033. Sidecar variants compare...
SRC00049	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0033. Sidecar variants compare...
SRC00050	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0020. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00051	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0020. Sidecar variants compare...
SRC00052	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0020. Sidecar variants compare...
SRC00053	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0020. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00054	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0042. Sidecar variants compare...
SRC00055	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0042. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00056	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0042. Sidecar variants compare...
SRC00057	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0042. Sidecar variants compare...
SRC00058	0 covered · 0 excluded · 11 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...

Source	Unit counts	Example decisions
SRC00060	0 covered · 0 excluded · 11 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00059	0 covered · 0 excluded · 11 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00061	0 covered · 0 excluded · 7 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00063	0 covered · 0 excluded · 7 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00062	0 covered · 0 excluded · 7 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00064	6 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00066	6 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00065	6 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00067	0 covered · 0 excluded · 13 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00069	0 covered · 0 excluded · 13 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00068	0 covered · 0 excluded · 13 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00070	0 covered · 0 excluded · 11 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00072	0 covered · 0 excluded · 11 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00071	0 covered · 0 excluded · 11 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00073	0 covered · 0 excluded · 10 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00075	0 covered · 0 excluded · 10 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00074	0 covered · 0 excluded · 10 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00076	0 covered · 0 excluded · 17 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00078	0 covered · 0 excluded · 17 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00077	0 covered · 0 excluded · 17 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00079	0 covered · 0 excluded · 13 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00081	0 covered · 0 excluded · 13 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00080	0 covered · 0 excluded · 13 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00082	0 covered · 0 excluded · 17 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00084	0 covered · 0 excluded · 17 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00083	0 covered · 0 excluded · 17 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00085	0 covered · 0 excluded · 9 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00087	0 covered · 0 excluded · 9 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00086	0 covered · 0 excluded · 9 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00088	0 covered · 0 excluded · 21 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00090	0 covered · 0 excluded · 21 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00089	0 covered · 0 excluded · 21 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00091	4 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00093	4 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00092	4 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00094	1 covered · 0 excluded · 0 gaps	q12: Original archive PDF page 12; candidate unit association is text-based and unconfirmed.
SRC00095	1 covered · 0 excluded · 0 gaps	q2: Original archive PDF page 2; candidate unit association is text-based and unconfirmed.
SRC00096	1 covered · 0 excluded · 0 gaps	q2: Original archive PDF page 2; candidate unit association is text-based and unconfirmed.
SRC00097	1 covered · 0 excluded · 0 gaps	q1: Original archive PDF page 1; candidate unit association is text-based and unconfirmed.
SRC00098	1 covered · 0 excluded · 0 gaps	q1: Original archive PDF page 1; candidate unit association is text-based and unconfirmed.
SRC00099	2 covered · 0 excluded · 0 gaps	q1: Original archive PDF page 1; candidate unit association is text-based and unconfirmed. q2: Original archive PDF page 2; candidate unit association is text-based and unconfirmed.
SRC00100	1 covered · 0 excluded · 0 gaps	q1: Original archive PDF page 1; candidate unit association is text-based and unconfirmed.
SRC00101	1 covered · 0 excluded · 0 gaps	q1: Original archive PDF page 1; candidate unit association is text-based and unconfirmed.

Source	Unit counts	Example decisions
SRC00102	2 covered · 0 excluded · 0 gaps	q1: Original archive PDF page 1; candidate unit association is text-based and unconfirmed. q2: Original archive PDF page 2; candidate unit association is text-based and unconfirmed.
SRC00103	1 covered · 0 excluded · 0 gaps	q2: Original archive PDF page 2; candidate unit association is text-based and unconfirmed.
SRC00104	1 covered · 0 excluded · 0 gaps	q1: Original archive PDF page 1; candidate unit association is text-based and unconfirmed.

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